

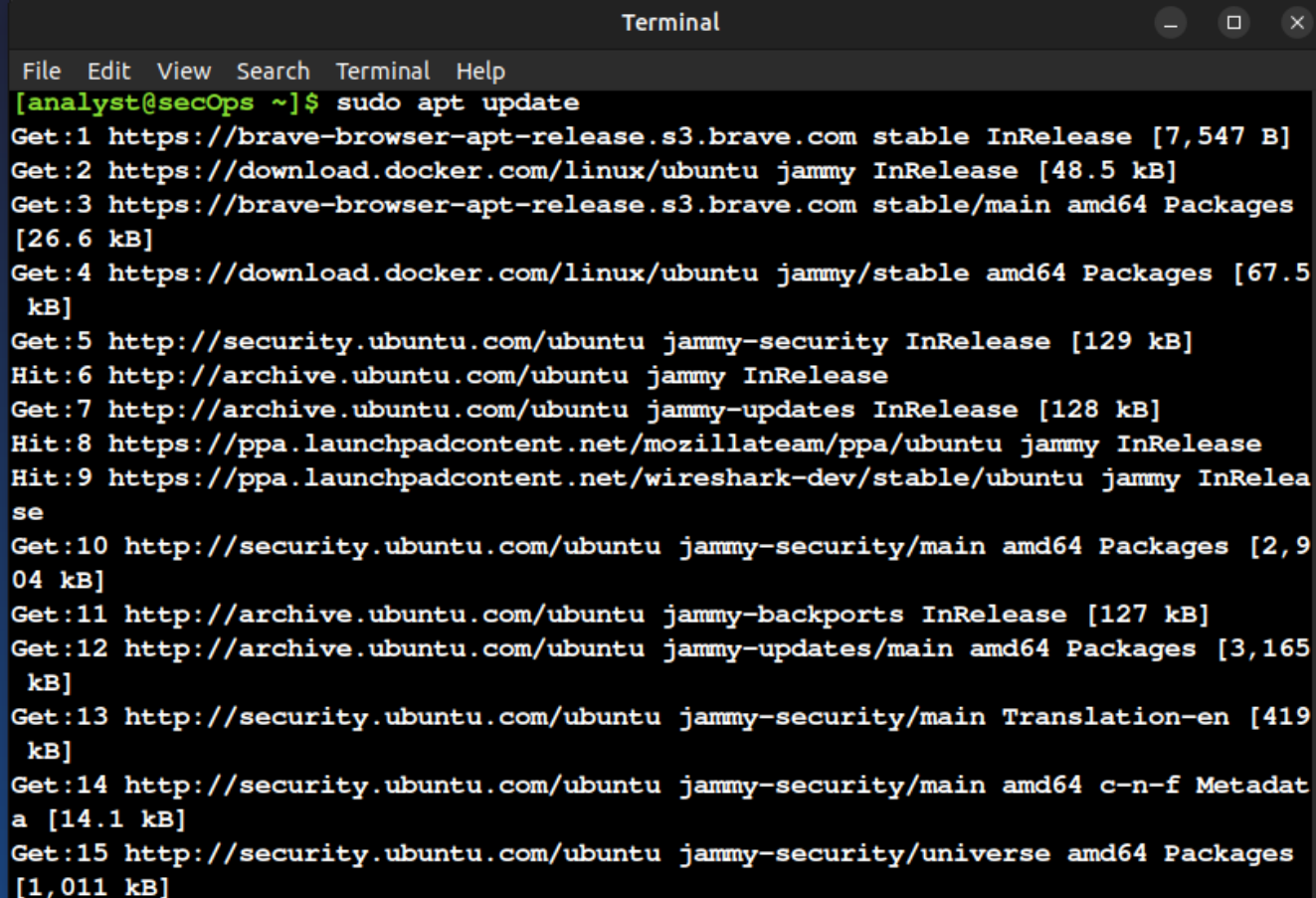
DNS Analysis & Troubleshooting Using Wireshark

Graded Lab Project
Ubuntu & Wireshark

Step 1

Command: **sudo apt update**

Explanation: Updates the package list and checks for available software updates.

A terminal window titled "Terminal" with a menu bar (File, Edit, View, Search, Terminal, Help) and window controls. The prompt is [analyst@secOps ~]\$ and the command executed is sudo apt update. The output shows 15 lines of status messages for various package sources, including brave-browser, docker, security.ubuntu.com, archive.ubuntu.com, and ppa.launchpadcontent.net.

```
Terminal
File Edit View Search Terminal Help
[analyst@secOps ~]$ sudo apt update
Get:1 https://brave-browser-apt-release.s3.brave.com stable InRelease [7,547 B]
Get:2 https://download.docker.com/linux/ubuntu jammy InRelease [48.5 kB]
Get:3 https://brave-browser-apt-release.s3.brave.com stable/main amd64 Packages
[26.6 kB]
Get:4 https://download.docker.com/linux/ubuntu jammy/stable amd64 Packages [67.5
kB]
Get:5 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:6 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Hit:8 https://ppa.launchpadcontent.net/mozillateam/ppa/ubuntu jammy InRelease
Hit:9 https://ppa.launchpadcontent.net/wireshark-dev/stable/ubuntu jammy InRelea
se
Get:10 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2,9
04 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3,165
kB]
Get:13 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [419
kB]
Get:14 http://security.ubuntu.com/ubuntu jammy-security/main amd64 c-n-f Metadat
a [14.1 kB]
Get:15 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages
[1,011 kB]
```

Step 2

Command: **sudo apt install wireshark -y**

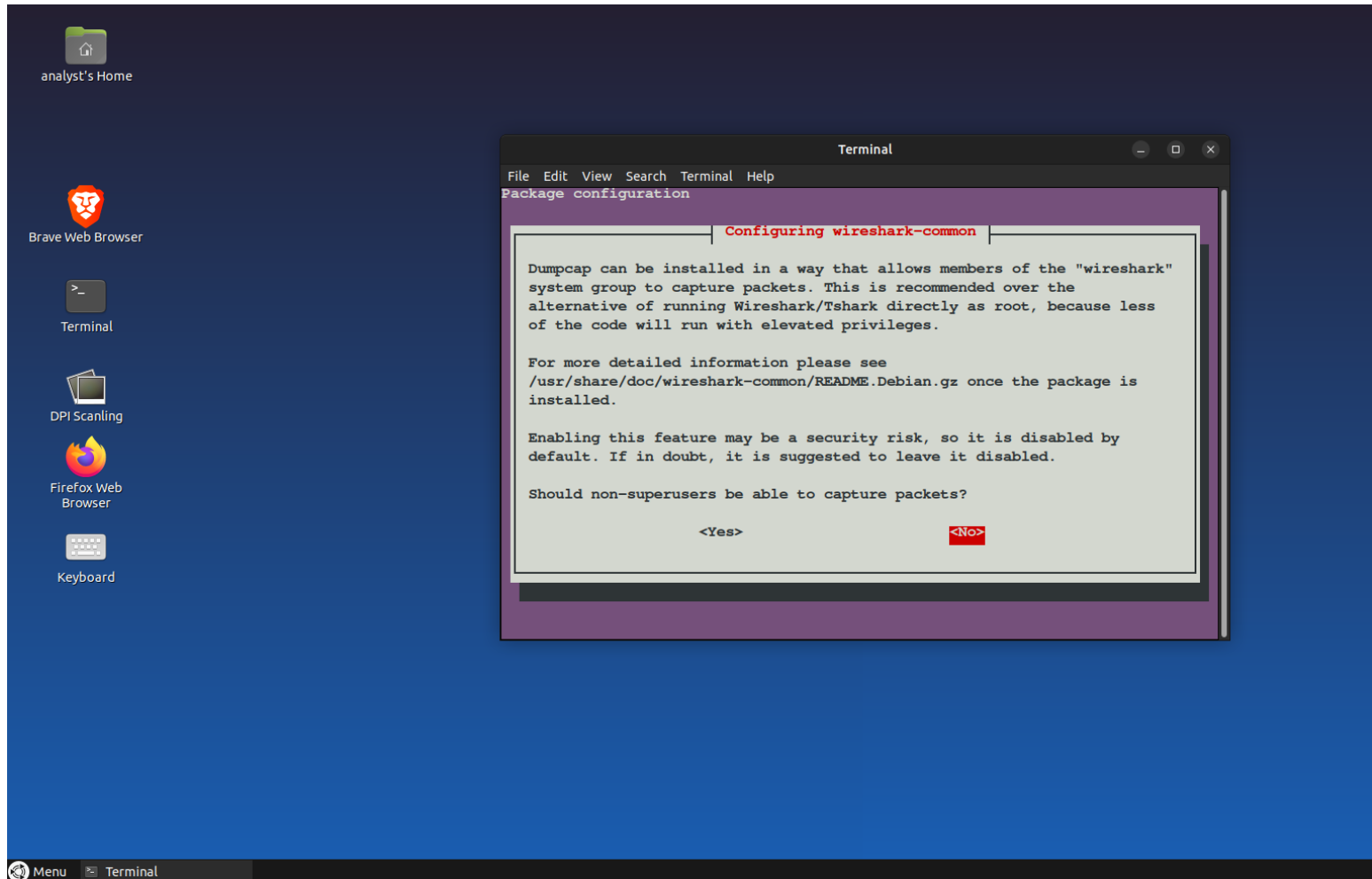
Explanation: Installs Wireshark on the Ubuntu system.

```
Terminal
File Edit View Search Terminal Help
[analyst@secOps ~]$ sudo apt install wireshark -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libbcbg729-0 libc-ares2 liblua5.4-0 libnghttp3-0 libopencore-amrnb0
  libqt5multimedia5 libqt5multimedia5-plugins libqt5multimediastools5
  libqt5multimediawidgets5 libqt5printrsupport5 libsbc1 libsmi2ldbl libspandsp2
  libwireshark-data libwireshark19 libwiretap16 libwsutil17 wireshark-common
Suggested packages:
  snmp-mibs-downloader geoipupdate geoip-database geoip-database-extra
  libjs-leaflet libjs-leaflet.markercluster wireshark-doc
The following NEW packages will be installed:
  libbcbg729-0 libc-ares2 liblua5.4-0 libnghttp3-0 libopencore-amrnb0
  libqt5multimedia5 libqt5multimedia5-plugins libqt5multimediastools5
  libqt5multimediawidgets5 libqt5printrsupport5 libsbc1 libsmi2ldbl libspandsp2
  libwireshark-data libwireshark19 libwiretap16 libwsutil17 wireshark
  wireshark-common
0 upgraded, 19 newly installed, 0 to remove and 421 not upgraded.
Need to get 33.1 MB of archives.
After this operation, 161 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libbcbg729-0 amd64 1.
1.1-2 [32.9 kB]
Get:2 https://ppa.launchpadcontent.net/wireshark-dev/stable/ubuntu jammy/main am
```

Step 3

Command: **Wireshark configuration prompt**

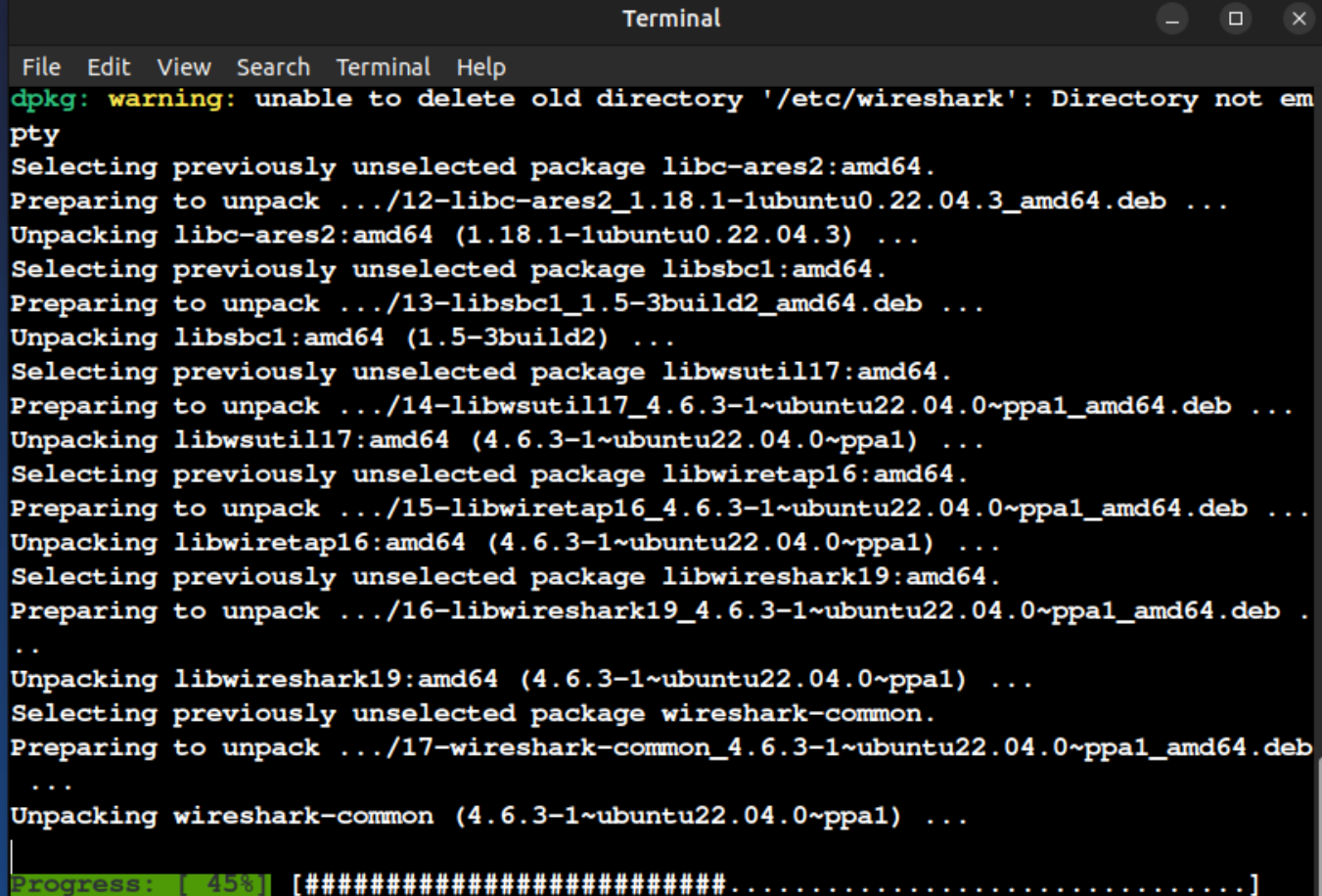
Explanation: Allows non-root users to capture packets by selecting Yes.



Step 4

Command: Installing dependencies

Explanation: Wireshark and required libraries are being installed.

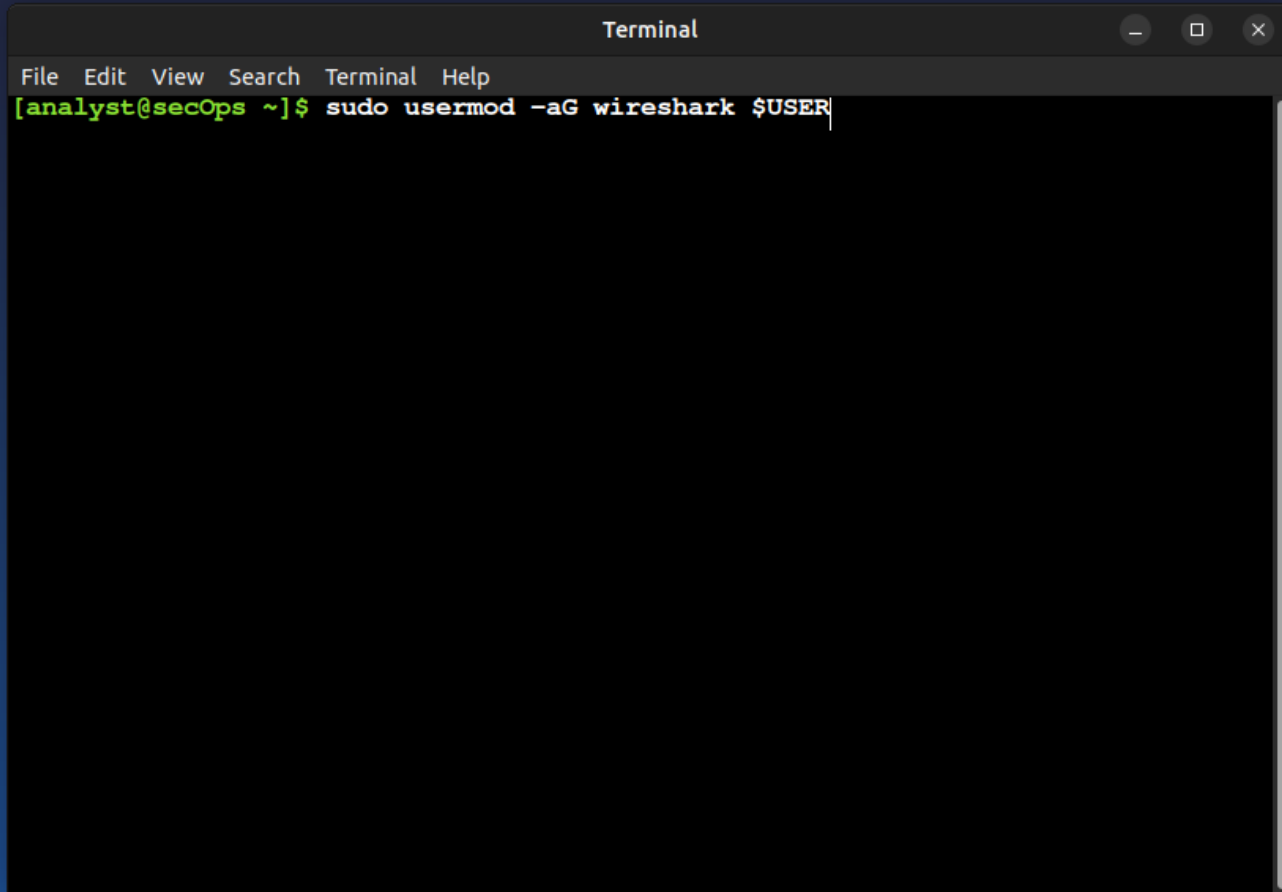
A terminal window titled "Terminal" with a menu bar (File, Edit, View, Search, Terminal, Help). The output shows the installation of several packages: libc-ares2:amd64, libsbc1:amd64, libwsutil17:amd64, libwiretap16:amd64, libwireshark19:amd64, and wireshark-common. A warning message at the top states: "dpkg: warning: unable to delete old directory '/etc/wireshark': Directory not empty". The progress bar at the bottom indicates 45% completion.

```
Terminal
File Edit View Search Terminal Help
dpkg: warning: unable to delete old directory '/etc/wireshark': Directory not empty
Selecting previously unselected package libc-ares2:amd64.
Preparing to unpack .../12-libc-ares2_1.18.1-1ubuntu0.22.04.3_amd64.deb ...
Unpacking libc-ares2:amd64 (1.18.1-1ubuntu0.22.04.3) ...
Selecting previously unselected package libsbc1:amd64.
Preparing to unpack .../13-libsbc1_1.5-3build2_amd64.deb ...
Unpacking libsbc1:amd64 (1.5-3build2) ...
Selecting previously unselected package libwsutil17:amd64.
Preparing to unpack .../14-libwsutil17_4.6.3-1~ubuntu22.04.0~ppa1_amd64.deb ...
Unpacking libwsutil17:amd64 (4.6.3-1~ubuntu22.04.0~ppa1) ...
Selecting previously unselected package libwiretap16:amd64.
Preparing to unpack .../15-libwiretap16_4.6.3-1~ubuntu22.04.0~ppa1_amd64.deb ...
Unpacking libwiretap16:amd64 (4.6.3-1~ubuntu22.04.0~ppa1) ...
Selecting previously unselected package libwireshark19:amd64.
Preparing to unpack .../16-libwireshark19_4.6.3-1~ubuntu22.04.0~ppa1_amd64.deb ...
Unpacking libwireshark19:amd64 (4.6.3-1~ubuntu22.04.0~ppa1) ...
Selecting previously unselected package wireshark-common.
Preparing to unpack .../17-wireshark-common_4.6.3-1~ubuntu22.04.0~ppa1_amd64.deb ...
Unpacking wireshark-common (4.6.3-1~ubuntu22.04.0~ppa1) ...
Progress: [ 45%] [#####.....]
```

Step 5

Command: **sudo usermod -aG wireshark \$USER**

Explanation: Adds the current user to the wireshark group.

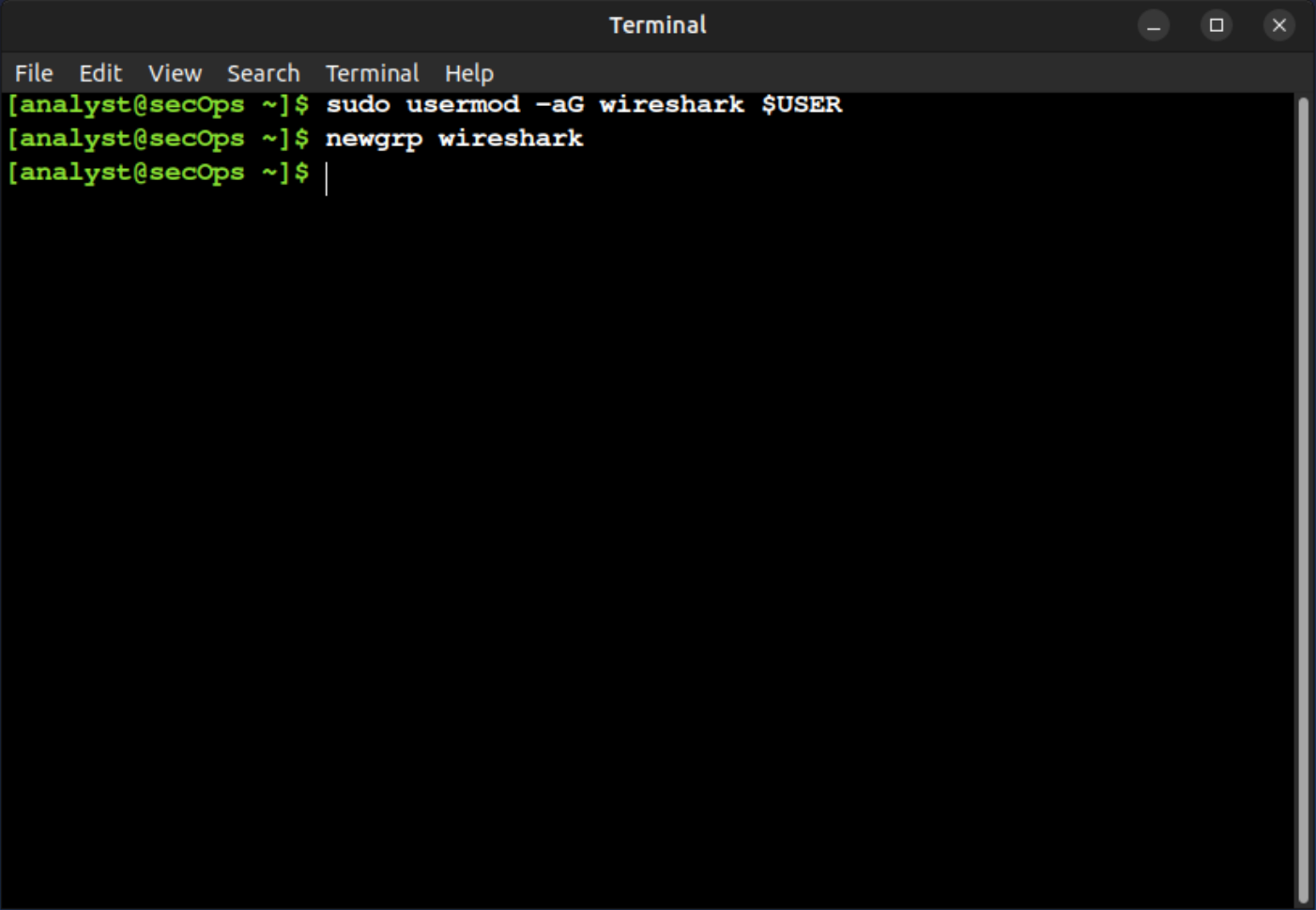
A screenshot of a terminal window titled "Terminal". The window has a menu bar with "File", "Edit", "View", "Search", "Terminal", and "Help". The terminal prompt is "[analyst@secOps ~]\$". The command "sudo usermod -aG wireshark \$USER" is being entered at the prompt. The terminal background is black, and the text is green. The window is set against a dark blue background.

```
Terminal
File Edit View Search Terminal Help
[analyst@secOps ~]$ sudo usermod -aG wireshark $USER
```

Step 6

Command: **newgrp wireshark**

Explanation: Applies the group change without logging out.

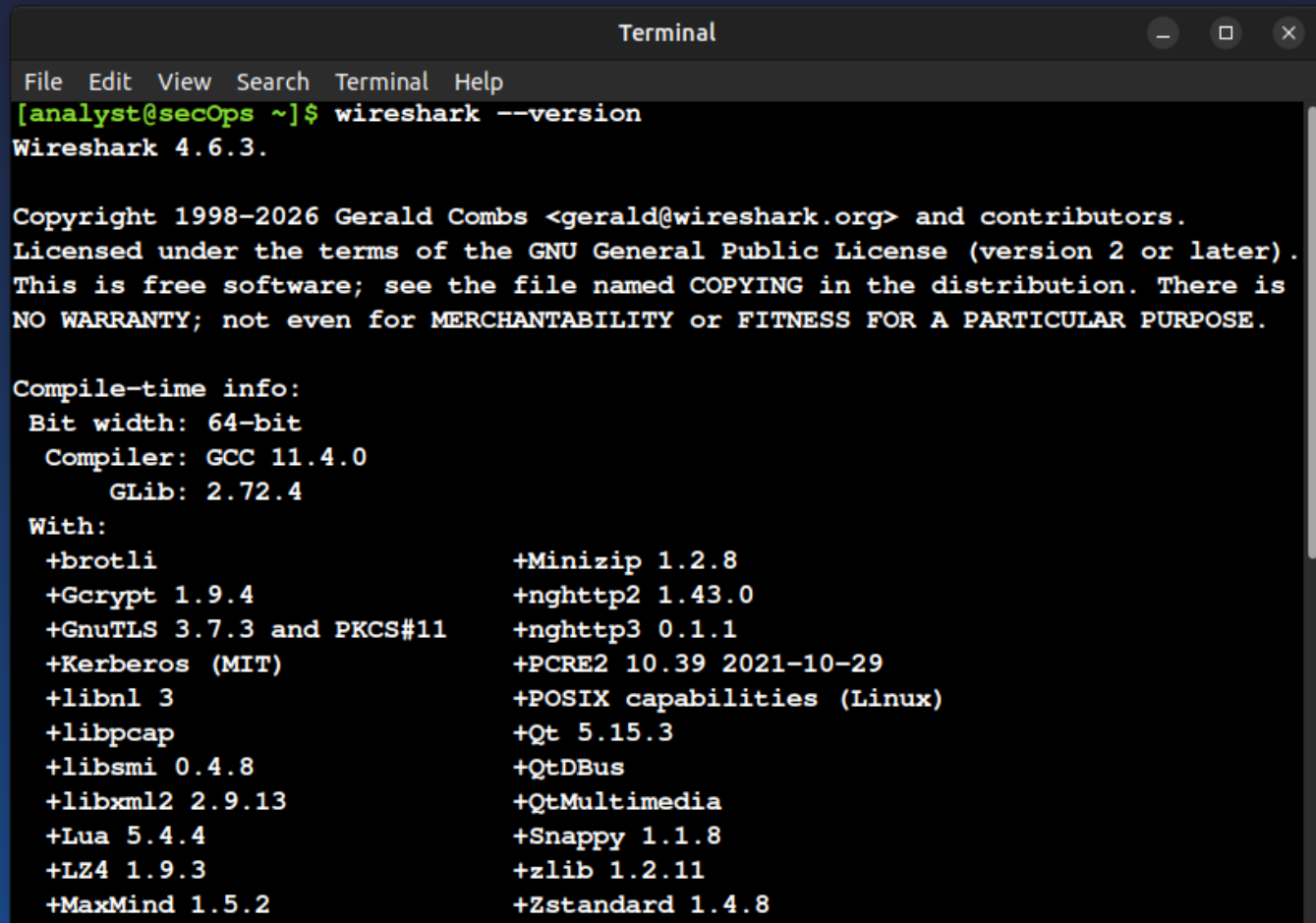


```
Terminal
File Edit View Search Terminal Help
[analyst@secOps ~]$ sudo usermod -aG wireshark $USER
[analyst@secOps ~]$ newgrp wireshark
[analyst@secOps ~]$
```

Step 7

Command: **wireshark --version**

Explanation: Verifies that Wireshark is installed correctly.

A terminal window titled "Terminal" with a menu bar (File, Edit, View, Search, Terminal, Help). The prompt is [analyst@secOps ~]\$ and the command entered is wireshark --version. The output shows the version (4.6.3), copyright information (1998-2026 Gerald Combs), license (GNU GPL), and a detailed list of compile-time dependencies and their versions.

```
Terminal
File Edit View Search Terminal Help
[analyst@secOps ~]$ wireshark --version
Wireshark 4.6.3.

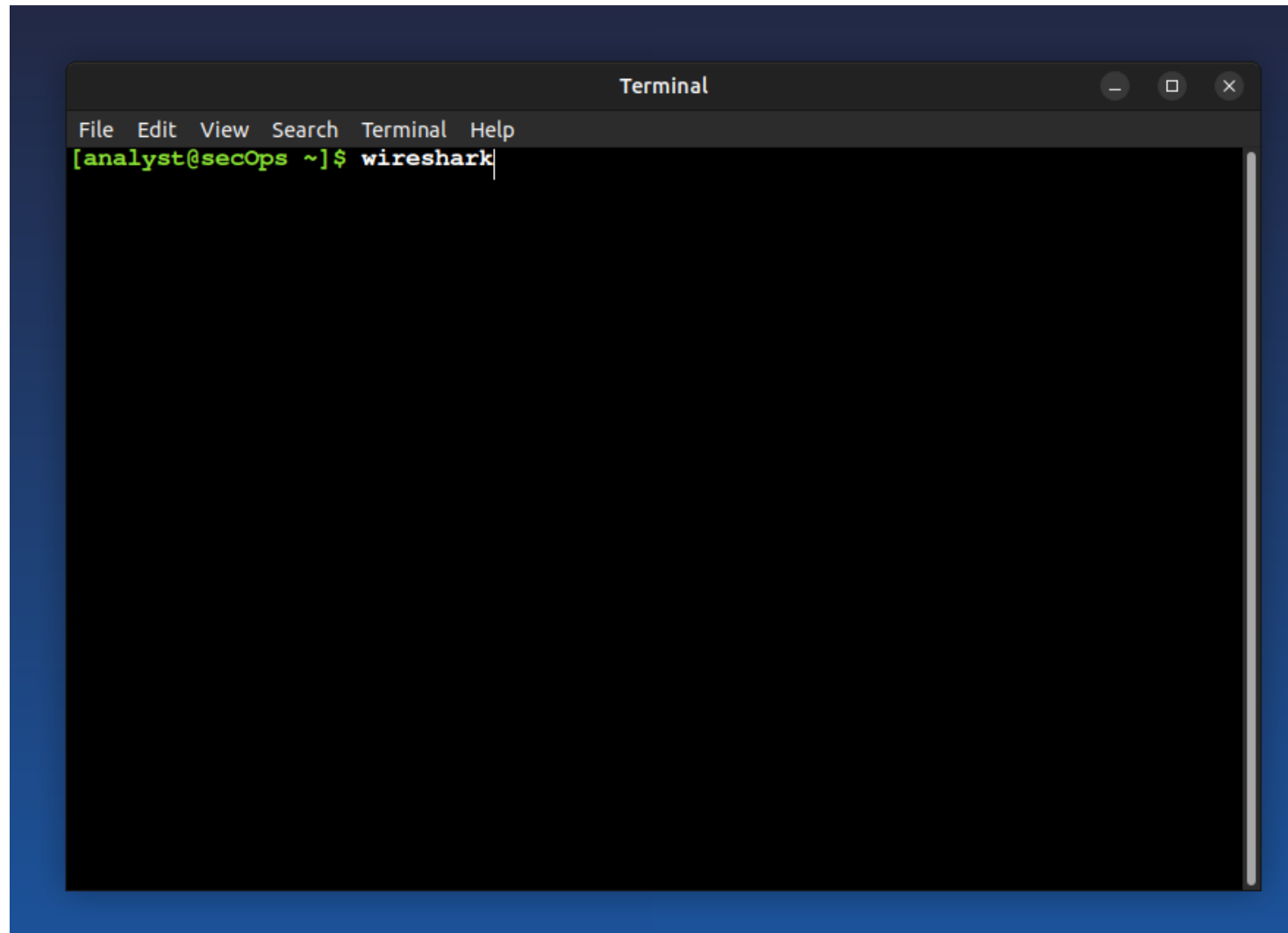
Copyright 1998-2026 Gerald Combs <gerald@wireshark.org> and contributors.
Licensed under the terms of the GNU General Public License (version 2 or later).
This is free software; see the file named COPYING in the distribution. There is
NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compile-time info:
  Bit width: 64-bit
  Compiler: GCC 11.4.0
  GLib: 2.72.4
  With:
    +brotli
    +Gcrypt 1.9.4
    +GnuTLS 3.7.3 and PKCS#11
    +Kerberos (MIT)
    +libnl 3
    +libpcap
    +libsmi 0.4.8
    +libxml2 2.9.13
    +Lua 5.4.4
    +LZ4 1.9.3
    +MaxMind 1.5.2
    +Minizip 1.2.8
    +nghttp2 1.43.0
    +nghttp3 0.1.1
    +PCRE2 10.39 2021-10-29
    +POSIX capabilities (Linux)
    +Qt 5.15.3
    +QtDBus
    +QtMultimedia
    +Snappy 1.1.8
    +zlib 1.2.11
    +Zstandard 1.4.8
```


Step 8

Command: **wireshark**

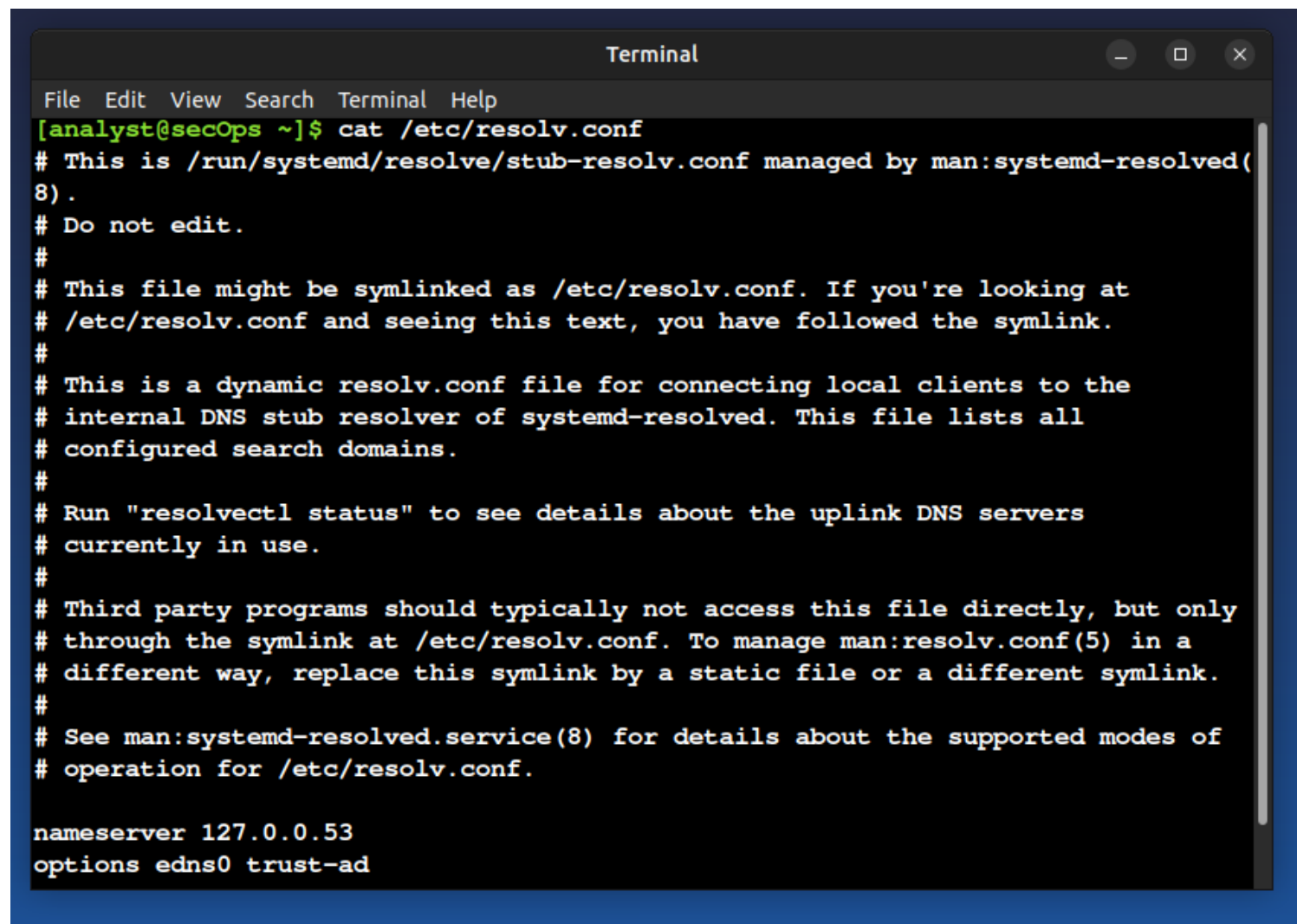
Explanation: Launches the Wireshark application.



Step 9

Command: **cat /etc/resolv.conf**

Explanation: Shows the current DNS configuration used by the system.

A terminal window titled "Terminal" with standard window controls (minimize, maximize, close) in the top right corner. The terminal has a menu bar with "File", "Edit", "View", "Search", "Terminal", and "Help". The prompt is "[analyst@secOps ~]\$". The command "cat /etc/resolv.conf" has been executed, displaying the contents of the file. The output consists of several lines of comments starting with "#", followed by the DNS configuration lines "nameserver 127.0.0.53" and "options edns0 trust-ad".

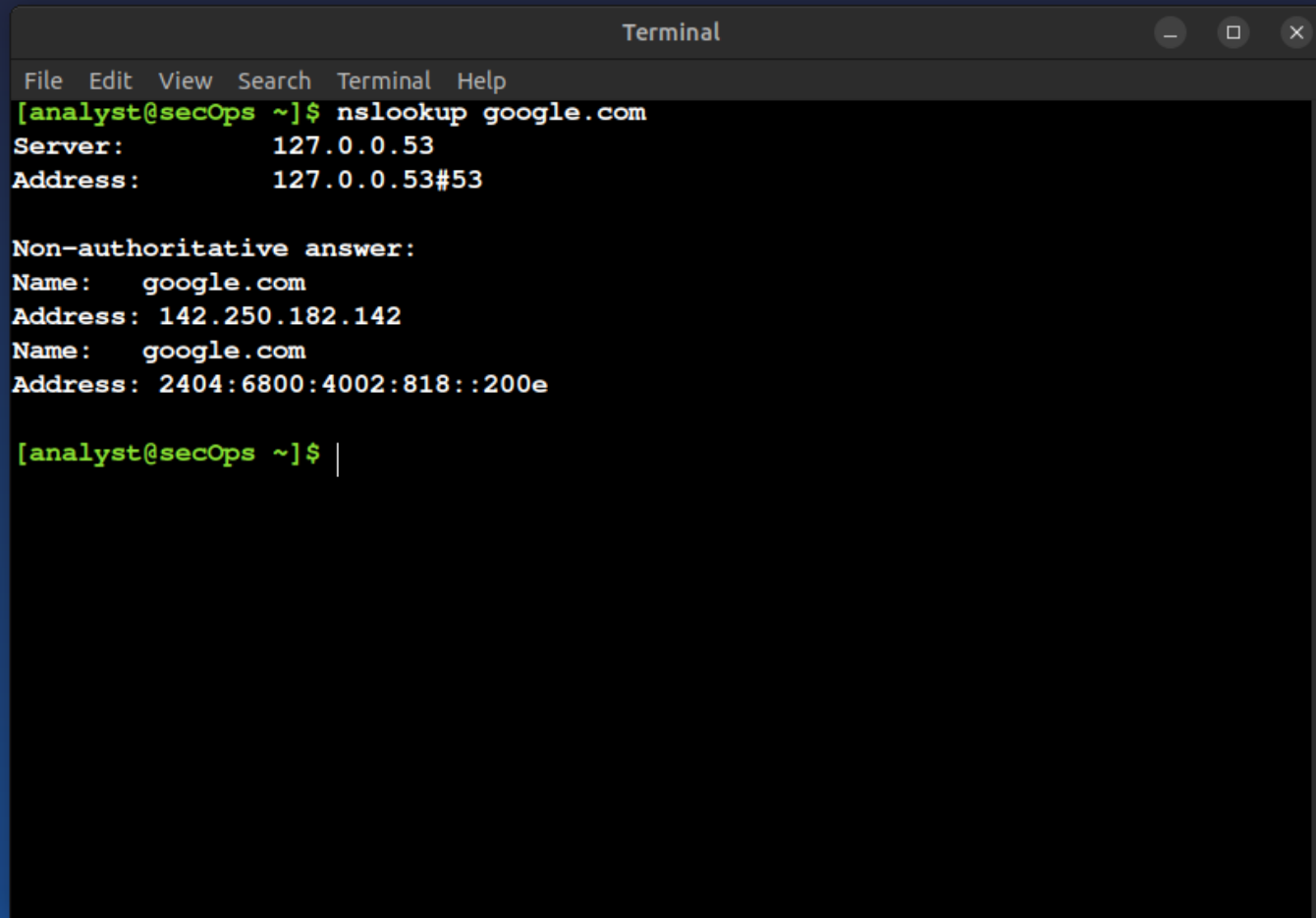
```
Terminal
File Edit View Search Terminal Help
[analyst@secOps ~]$ cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(
8) .
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 127.0.0.53
options edns0 trust-ad
```

Step 10

Command: **nslookup google.com**

Explanation: Tests DNS resolution for a valid domain.

A screenshot of a macOS Terminal window titled "Terminal". The window has a menu bar with "File", "Edit", "View", "Search", "Terminal", and "Help". The prompt is "[analyst@secOps ~]\$". The command "nslookup google.com" has been entered. The output shows the server and address of the local DNS resolver, followed by a "Non-authoritative answer:" section containing the IPv4 and IPv6 addresses for google.com. The prompt is now "[analyst@secOps ~]\$ |".

```
Terminal
File Edit View Search Terminal Help
[analyst@secOps ~]$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

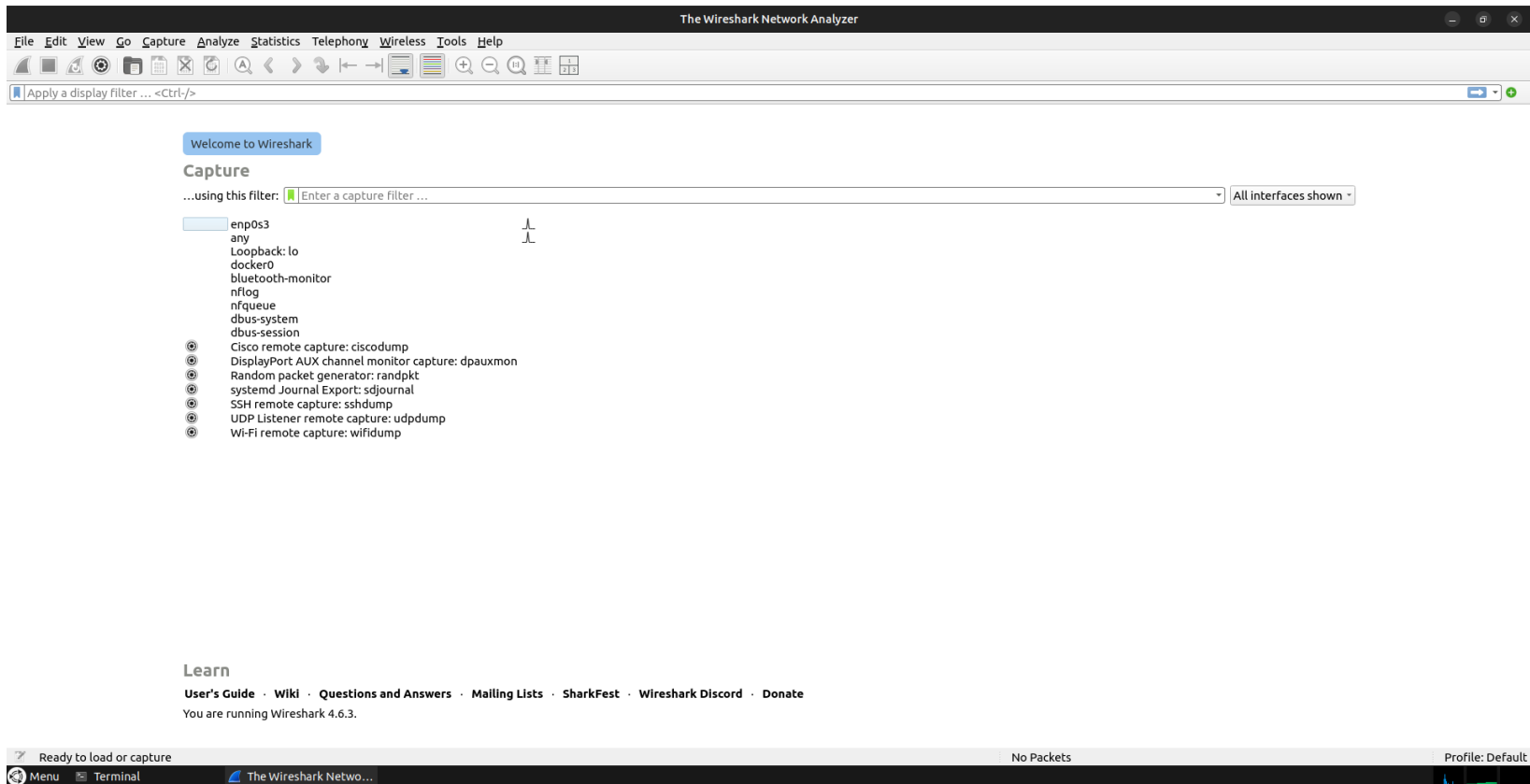
Non-authoritative answer:
Name:   google.com
Address: 142.250.182.142
Name:   google.com
Address: 2404:6800:4002:818::200e

[analyst@secOps ~]$ |
```

Step 11

Command: **Open Wireshark**

Explanation: Wireshark interface is opened for packet capture.



Step 12

Command: **Apply dns filter**

Explanation: Filters only DNS packets in Wireshark.

The screenshot shows the Wireshark interface with the filter 'dns' applied to the packet list. The packet list displays several DNS and NTP packets. The packet details pane shows the structure of a Network Time Protocol (NTP) Version 4, server packet.

No.	Source	Destination	Protocol	Length	Info
1	10.0.2.15	139.59.55.93	NTP	90	NTP Version 4, client
2	139.59.55.93	10.0.2.15	NTP	90	NTP Version 4, server
3	PCSSystemtec_ed:74:...	52:55:0a:00:02:02	ARP	42	Who has 10.0.2.2? Tell 10.0.2.15
4	52:55:0a:00:02:02	PCSSystemtec_ed:74:...	ARP	64	10.0.2.2 is at 52:55:0a:00:02:02
5	10.0.2.15	95.216.192.15	NTP	90	NTP Version 4, client
6	95.216.192.15	10.0.2.15	NTP	90	NTP Version 4, server
7	10.0.2.15	65.0.119.56	NTP	90	NTP Version 4, client
8	65.0.119.56	10.0.2.15	NTP	90	NTP Version 4, server

Frame 2: Packet, 90 bytes on wire (720 bits), 90 bytes captured (720 bits) on interface enp0s3, id 0
Ethernet II, Src: 52:55:0a:00:02:02 (52:55:0a:00:02:02), Dst: PCSSystemtec_ed:74:19 (08:00:27:ed:74:19)
Internet Protocol Version 4, Src: 139.59.55.93, Dst: 10.0.2.15
User Datagram Protocol, Src Port: 123, Dst Port: 123
Network Time Protocol (NTP Version 4, server)

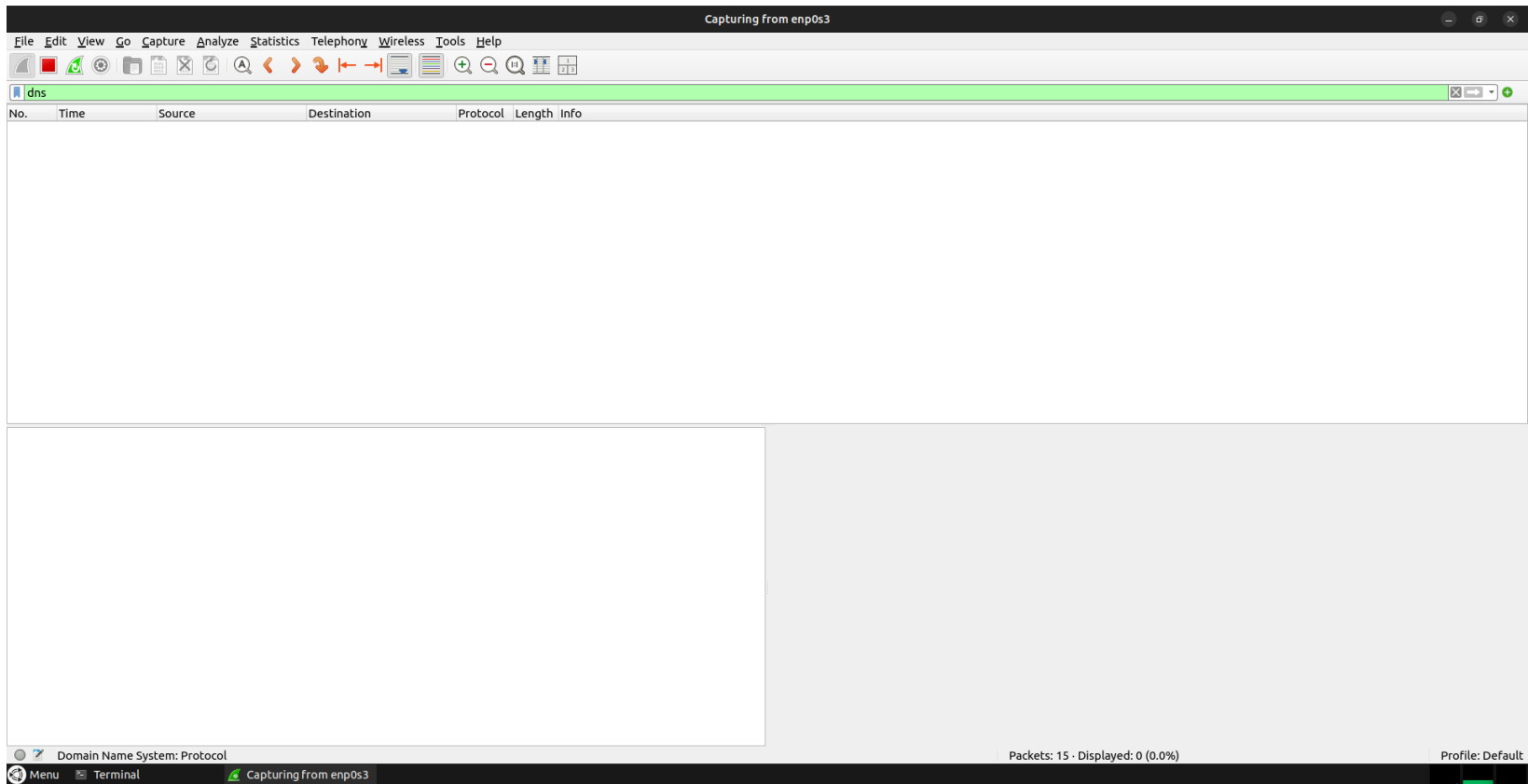
0000 08 00 27 ed 74 19 52 55 0a 00 02 02 08 00 45 00 ...t RUE
0010 00 4c bf 5e 00 00 40 11 ec 9b 8b 3b 37 5d 0a 00 ...L^..@...;7]..
0020 02 0f 00 7b 00 7b 00 38 91 52 24 02 06 e7 00 00 ...{.8.R\$.
0030 08 19 00 00 7b 0e 8b 3c 6a ed 1e bf 85 0d 5b ...{...<]....[
0040 43 e2 ed 1e c2 52 40 4b cb 91 ed 1e c2 52 49 49 C...R&K.....RII
0050 89 5b ed 1e c2 52 49 92 eb d0 [...RI..

Domain Name System: Protocol
Packets: 8
Profile: Default

Step 13

Command: **Start capture**

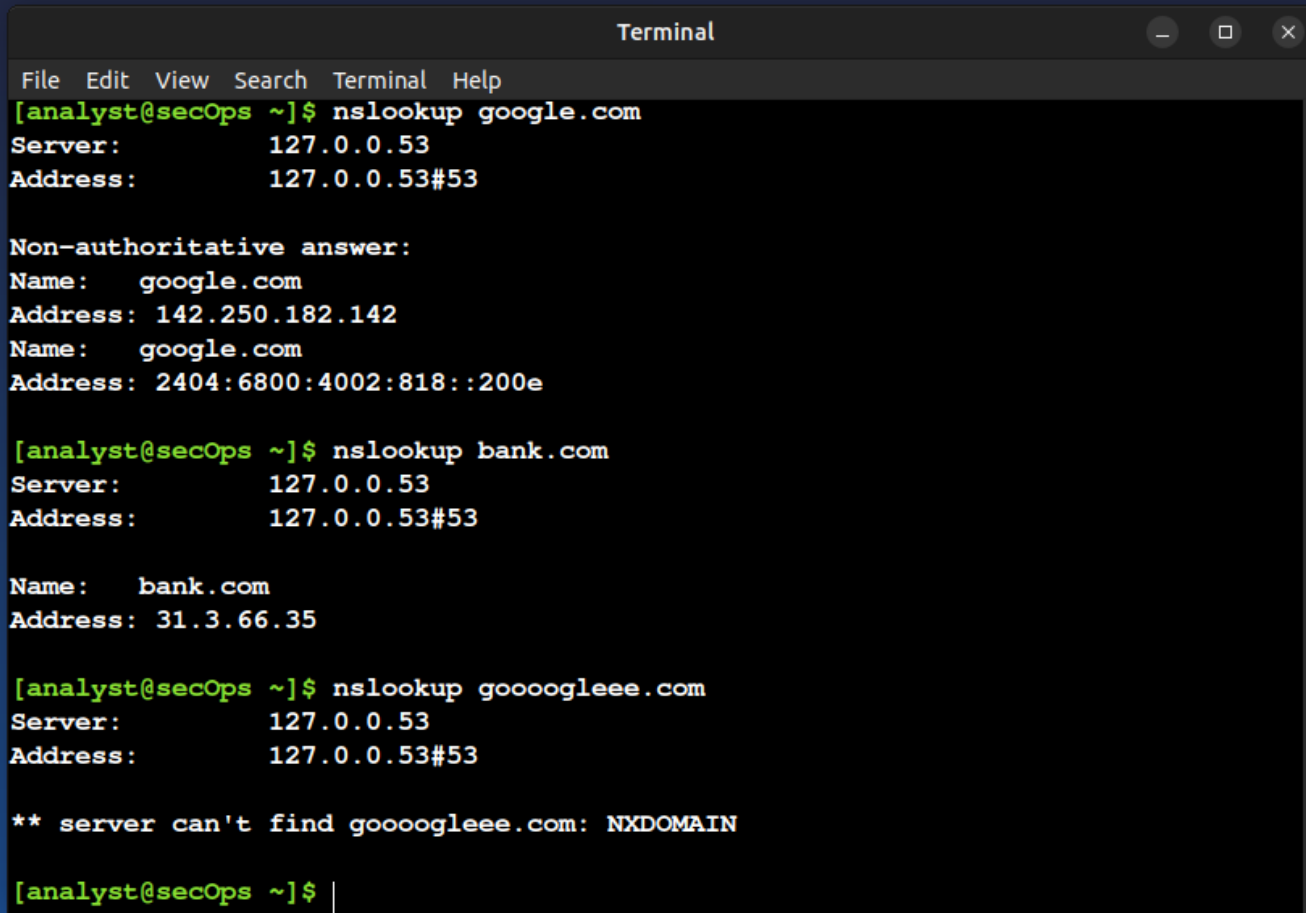
Explanation: Wireshark starts capturing DNS traffic.



Step 14

Command: **Generate DNS traffic**

Explanation: DNS queries are generated using nslookup.



```
Terminal
File Edit View Search Terminal Help
[analyst@secOps ~]$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.182.142
Name:   google.com
Address: 2404:6800:4002:818::200e

[analyst@secOps ~]$ nslookup bank.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Name:   bank.com
Address: 31.3.66.35

[analyst@secOps ~]$ nslookup goooogleee.com
Server:      127.0.0.53
Address:     127.0.0.53#53

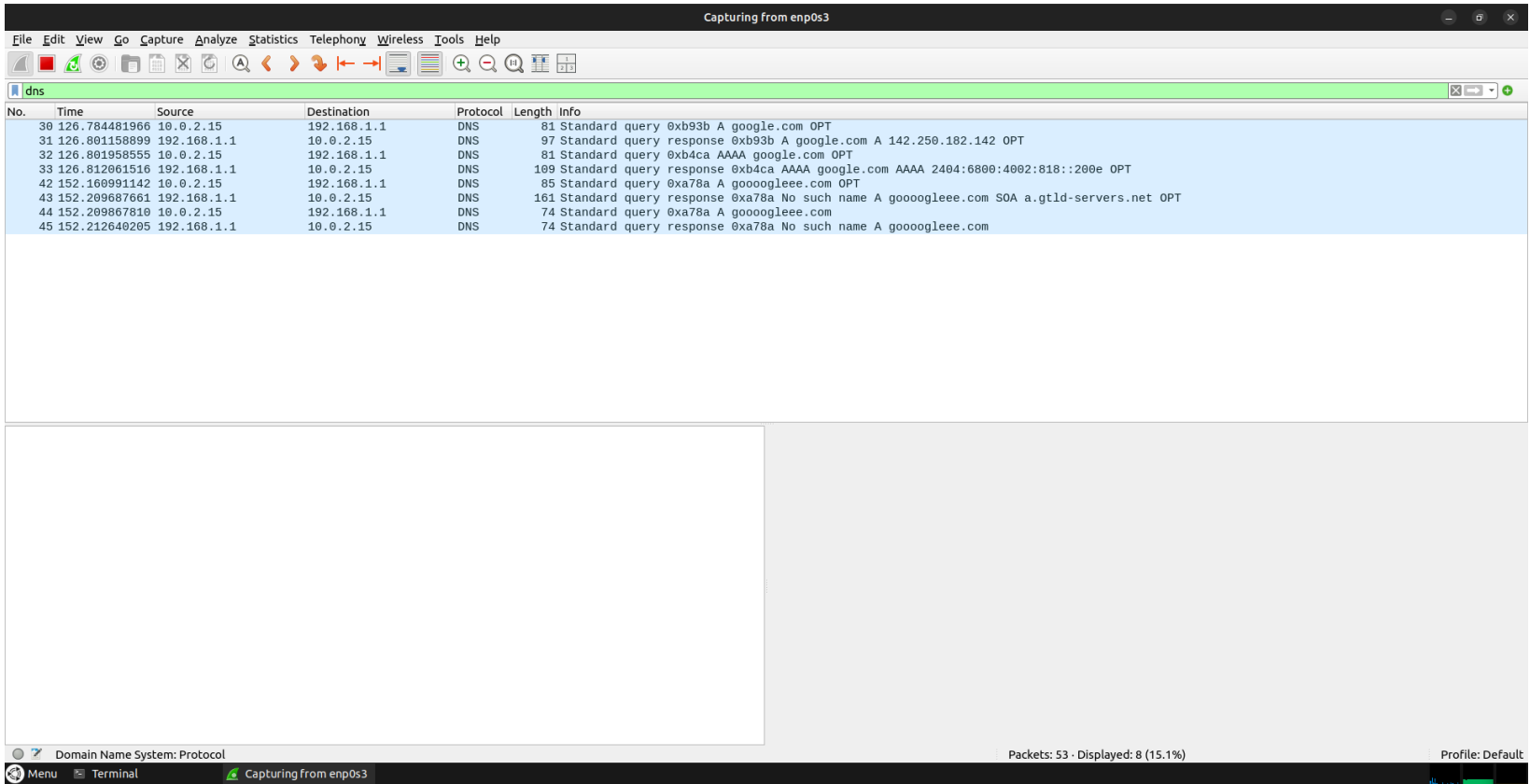
** server can't find goooogleee.com: NXDOMAIN

[analyst@secOps ~]$ |
```

Step 15

Command: **Analyze DNS packets**

Explanation: DNS requests and responses are visible in Wireshark.



The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for file operations, packet navigation, and analysis. The main window is titled "Capturing from enp0s3" and shows a list of captured packets filtered by the "dns" protocol. The packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
30	126.784481966	10.0.2.15	192.168.1.1	DNS	81	Standard query 0xb93b A google.com OPT
31	126.801158899	192.168.1.1	10.0.2.15	DNS	97	Standard query response 0xb93b A google.com A 142.250.182.142 OPT
32	126.801958555	10.0.2.15	192.168.1.1	DNS	81	Standard query 0xb4ca AAAA google.com OPT
33	126.812061516	192.168.1.1	10.0.2.15	DNS	109	Standard query response 0xb4ca AAAA google.com AAAA 2404:6800:4002:818::200e OPT
42	152.160991142	10.0.2.15	192.168.1.1	DNS	85	Standard query 0xa78a A goooogleeee.com OPT
43	152.209687661	192.168.1.1	10.0.2.15	DNS	161	Standard query response 0xa78a No such name A goooogleeee.com SOA a.gtld-servers.net OPT
44	152.209867810	10.0.2.15	192.168.1.1	DNS	74	Standard query 0xa78a A goooogleeee.com
45	152.212640205	192.168.1.1	10.0.2.15	DNS	74	Standard query response 0xa78a No such name A goooogleeee.com

The bottom status bar indicates "Packets: 53 · Displayed: 8 (15.1%)" and "Profile: Default". The bottom-most bar shows "Menu", "Terminal", and "Capturing from enp0s3".

Step 16

Command: Check response time (**google.com**)

Explanation: DNS response is fast due to reachable DNS server.

The image shows a Wireshark packet capture window titled "Capturing from enp0s3". The main packet list on the left shows several DNS packets. Packet 30 is selected, showing a "Standard query" for "A google.com OPT". The packet details pane on the right shows the structure of the DNS packet, including the section number, interface name, encapsulation type, arrival time, and frame length. The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII.

Wireshark - Packet 30 - enp0s3

Section number: 1

Interface id: 0 (enp0s3)

Interface name: enp0s3

Encapsulation type: Ethernet (1)

Arrival Time: Jan 24, 2026 04:02:25.035679248 UTC

UTC Arrival Time: Jan 24, 2026 04:02:25.035679248 UTC

Epoch Arrival Time: 1769227345.035679248

[Time shift for this packet: 0.00000000 seconds]

[Time delta from previous captured frame: 19.760046150 seconds]

[Time since reference or first frame: 2 minutes, 6.784481966 seconds]

Frame Number: 30

Frame Length: 81 bytes (648 bits)

Capture Length: 81 bytes (648 bits)

[Frame is marked: False]

Time delta from previous captured frame (frame.time_delta)

✓ Show packet bytes Layout: Vertical (Stacked)

Help Close

Packets: 84 · Displayed: 8 (9.5%) Profile: Default

Step 17

Command: Check **NXDOMAIN**

Explanation: Invalid domain returns NXDOMAIN response.

The image shows a Wireshark packet capture window titled "Capturing from enp0s3". The main packet list displays several DNS queries and responses. Packet 42 is highlighted, showing a DNS Standard query response for "google.com" with a status of "No such name".

No.	Time	Source	Destination	Protocol	Length	Info
30	126.784481966	10.0.2.15	192.168.1.1	DNS	81	Standard query 0xb93b A google.com OPT
31	126.801158899	192.168.1.1	10.0.2.15	DNS	97	Standard query response 0xb93b A google.com A 142.250.182.142 OPT
32	126.801958555	10.0.2.15	192.168.1.1	DNS	81	Standard query 0xb4ca AAAA google.com OPT
33	126.812061516	192.168.1.1	10.0.2.15	DNS	109	Standard query response 0xb4ca AAAA google.com AAAA 2404:6800:4002:818::200e OPT
42	152.160991142	10.0.2.15	192.168.1.1	DNS	85	Standard query 0xa78a A gooogoleee.com OPT
43	152.209687661	192.168.1.1	10.0.2.15	DNS	161	Standard query response 0xa78a No such name A gooogoleee.com SOA a.gtld-servers.net OPT
44	152.209867810	10.0.2.15	192.168.1.1	DNS	74	Standard query 0xa78a A gooogoleee.com
45	152.212640205	192.168.1.1	10.0.2.15	DNS	74	Standard query response 0xa78a No such name A gooogoleee.com

The packet details pane for packet 42 shows the following information:

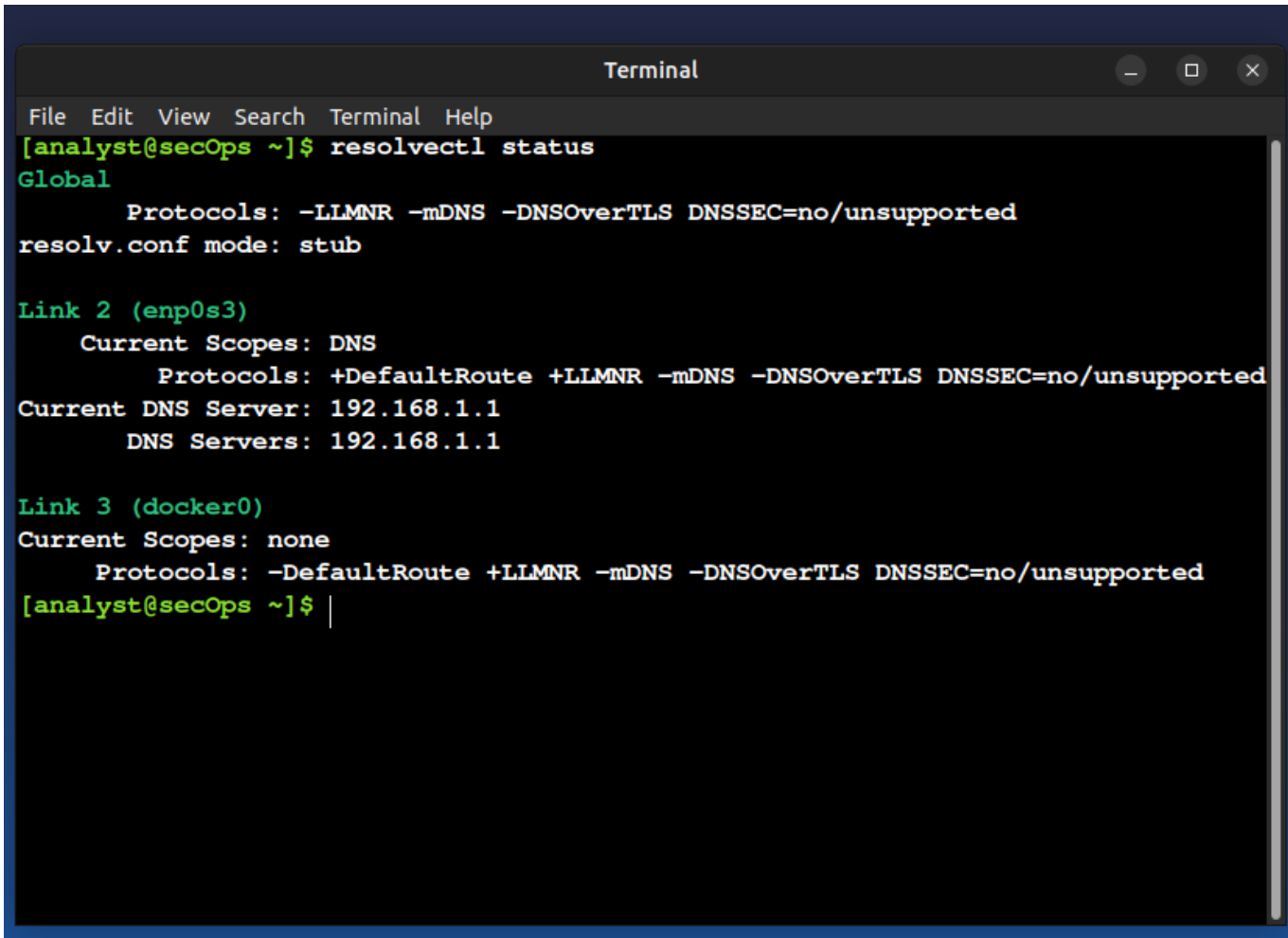
- Section number: 1
- Interface id: 0 (enp0s3)
- Interface name: enp0s3
- Encapsulation type: Ethernet (1)
- Arrival Time: Jan 24, 2026 04:02:50.412188424 UTC
- UTC Arrival Time: Jan 24, 2026 04:02:50.412188424 UTC
- Epoch Arrival Time: 1769227370.412188424
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 7.122340676 seconds]
- [Time delta from previous displayed frame: 25.348929626 seconds]
- [Time since reference or first frame: 2 minutes, 32.160991142 seconds]
- Frame Number: 42
- Frame Length: 85 bytes (680 bits)
- Capture Length: 85 bytes (680 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: eth:ethertype:ip:udp:dns]
- Character encoding: ASCII (0)
- [Coloring Rule Name: UDP]
- [Coloring Rule String: udp]

The packet bytes pane shows the raw data of the packet, including the DNS response structure.

Step 18

Command: **resolvectl status**

Explanation: Shows current DNS server configuration.

A terminal window titled "Terminal" with standard window controls (minimize, maximize, close) in the top right corner. The menu bar includes "File", "Edit", "View", "Search", "Terminal", and "Help". The prompt is "[analyst@secOps ~]\$". The command "resolvectl status" has been executed, showing the following output:

```
Global
  Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
resolv.conf mode: stub

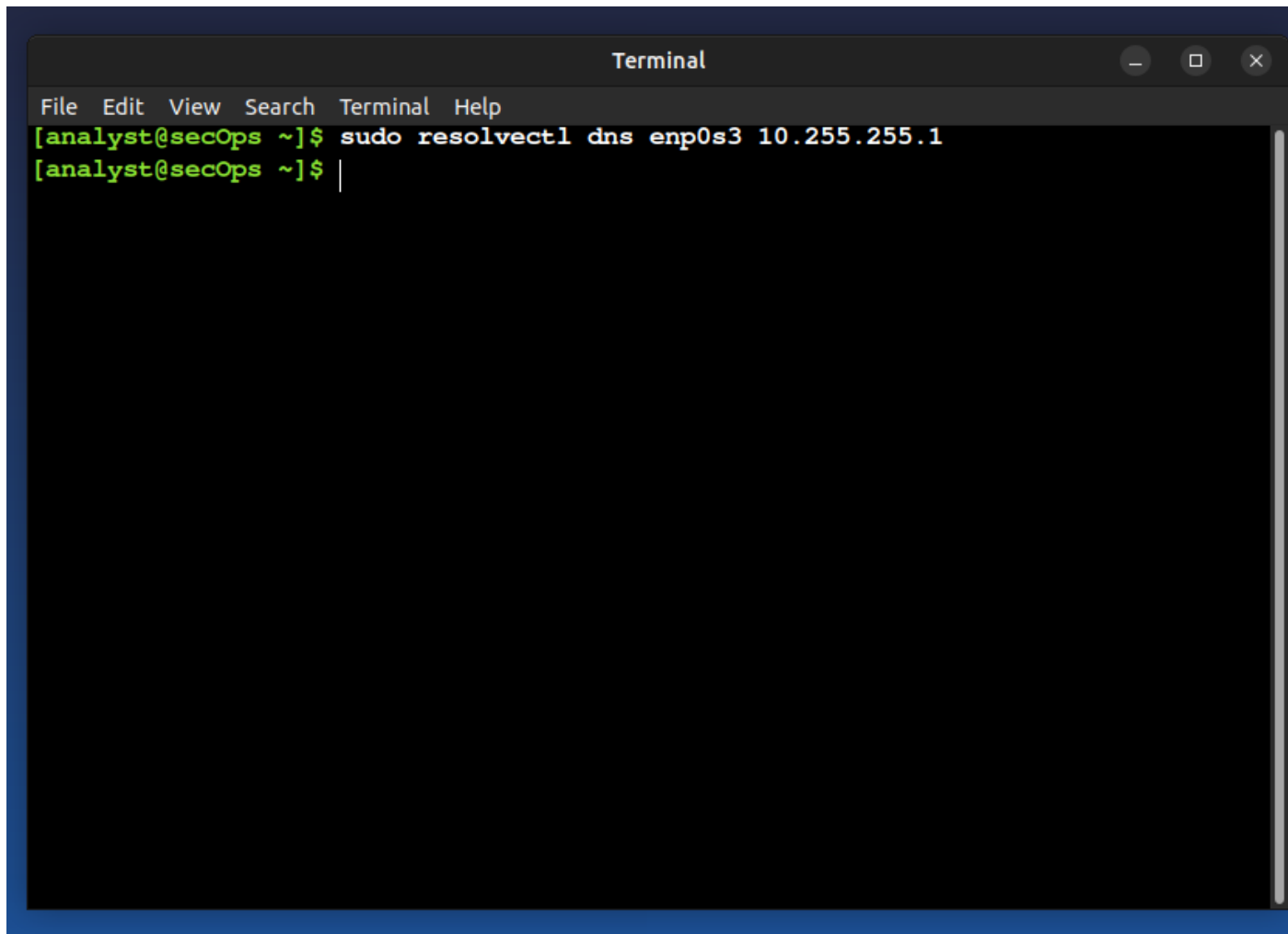
Link 2 (enp0s3)
  Current Scopes: DNS
    Protocols: +DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
Current DNS Server: 192.168.1.1
  DNS Servers: 192.168.1.1

Link 3 (docker0)
Current Scopes: none
  Protocols: -DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
[analyst@secOps ~]$ |
```

Step 19

Command: **Set bad DNS (10.255.255.1)**

Explanation: System DNS is changed to an unreachable server.

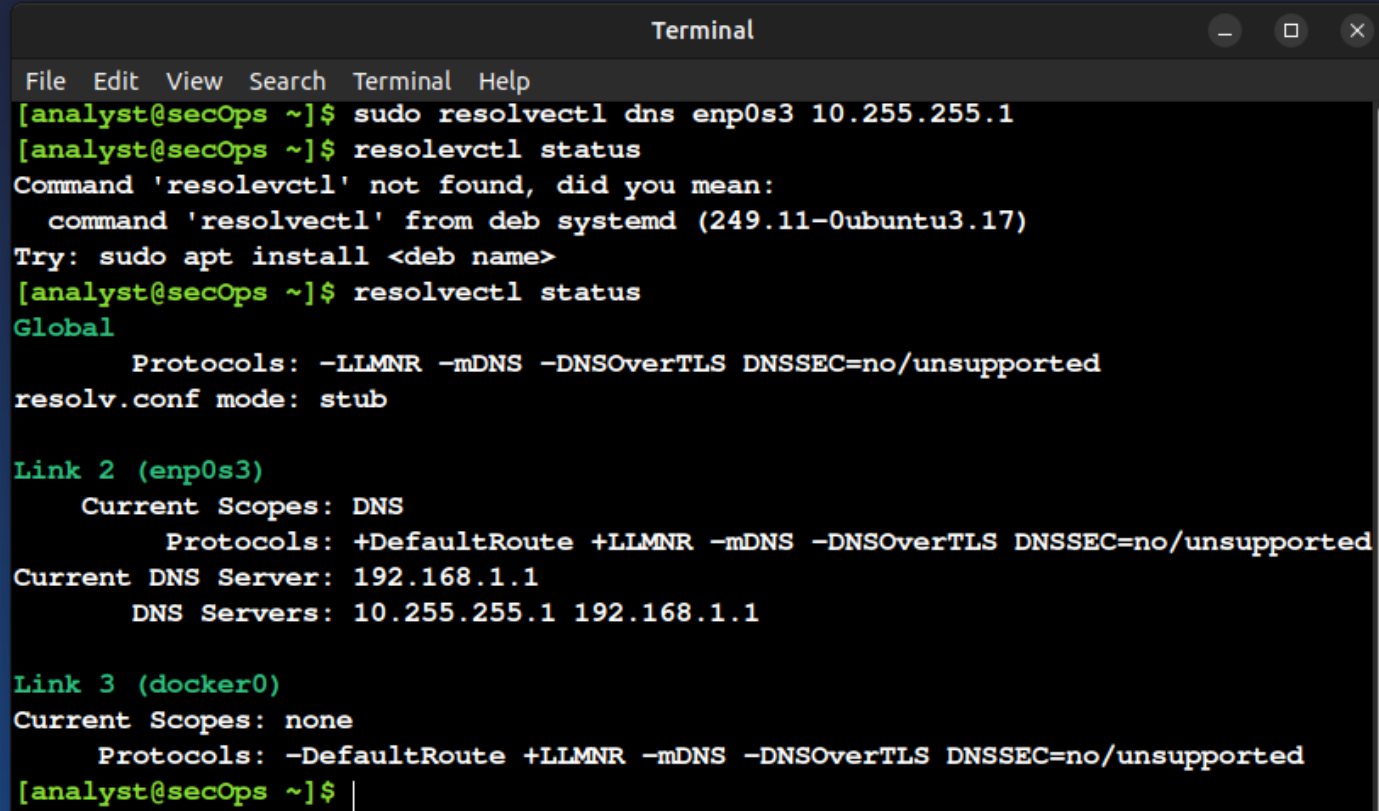
A terminal window titled "Terminal" with a dark background and a blue border. The window contains a menu bar with "File", "Edit", "View", "Search", "Terminal", and "Help". The prompt is "[analyst@secOps ~]\$". The command "sudo resolvectl dns enp0s3 10.255.255.1" has been entered and executed. The prompt is now "[analyst@secOps ~]\$" with a cursor at the end.

```
Terminal
File Edit View Search Terminal Help
[analyst@secOps ~]$ sudo resolvectl dns enp0s3 10.255.255.1
[analyst@secOps ~]$
```

Step 20

Test DNS again. Command: **resolvectl status**

Explanation: DNS resolution becomes slow due to timeout.



```
Terminal
File Edit View Search Terminal Help
[analyst@secOps ~]$ sudo resolvectl dns enp0s3 10.255.255.1
[analyst@secOps ~]$ resolvectl status
Command 'resolvectl' not found, did you mean:
  command 'resolvectl' from deb systemd (249.11-0ubuntu3.17)
Try: sudo apt install <deb name>
[analyst@secOps ~]$ resolvectl status
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
resolv.conf mode: stub

Link 2 (enp0s3)
    Current Scopes: DNS
        Protocols: +DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
Current DNS Server: 192.168.1.1
    DNS Servers: 10.255.255.1 192.168.1.1

Link 3 (docker0)
    Current Scopes: none
        Protocols: -DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
[analyst@secOps ~]$ |
```

Step 21

Observe delay

Explanation: Wireshark shows delayed DNS response.

In my ubuntu lab I noticed that the nslookup google.com took 2 seconds in terminal to show details and nslookup mygreatlearning.com took 4 to 5 seconds to show details in terminal.

The screenshot displays a Wireshark network capture window titled "Capturing from enp0s3". The DNS filter is applied, showing a list of captured packets. Packet 16 is selected, showing a standard query from 10.0.2.15 to 10.125.82.152. The packet details pane shows the Ethernet II header, IP header, and UDP header. The packet bytes pane shows the raw data. A terminal window is overlaid on the Wireshark window, showing the results of two nslookup commands. The first command is nslookup google.com, which returns the IP address 172.217.27.174. The second command is nslookup mygreatlearning.com, which returns the IP address 13.234.99.190. The terminal output shows the following:

```
[analyst@secOps ~]$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 172.217.27.174
Name:   google.com
Address: 2404:6800:4002:80e::200e

[analyst@secOps ~]$ nslookup mygreatlearning.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   mygreatlearning.com
Address: 13.234.99.190
Name:   mygreatlearning.com
Address: 3.6.246.229

[analyst@secOps ~]$
```

The Wireshark packet list shows the following details for packet 16:

No.	Time	Source	Destination	Protocol	Length	Info
16	1.065521245	10.0.2.15	10.125.82.152	DNS	81	Standard query 0x9bba to google.com OPT
17	0.036289846	10.125.82.152	10.0.2.15	DNS	81	Standard query response 0x9bba from 10.125.82.152 OPT
18	0.001749413	10.0.2.15	10.125.82.152	DNS	81	Standard query 0x9bba to google.com OPT
19	0.046802573	10.125.82.152	10.0.2.15	DNS	81	Standard query response 0x9bba from 10.125.82.152 OPT
20	2.812767876	10.0.2.15	10.125.82.152	DNS	81	Standard query 0x9bba to google.com OPT
23	0.000026359	fd17:625c:f037:2:a0...	fd17:625c:f037:2::3	DNS	81	Standard query 0x9bba to google.com OPT
24	0.220393078	fd17:625c:f037:2::3	fd17:625c:f037:2:a0...	DNS	81	Standard query response 0x9bba from fd17:625c:f037:2:a0...
25	0.001360013	fd17:625c:f037:2:a0...	fd17:625c:f037:2::3	DNS	81	Standard query 0x9bba to google.com OPT
26	0.056779481	fd17:625c:f037:2::3	fd17:625c:f037:2:a0...	DNS	81	Standard query response 0x9bba from fd17:625c:f037:2:a0...

The packet details pane for packet 16 shows the following information:

- Interface id: 0 (enp0s3)
- Interface name: enp0s3
- Encapsulation type: Ethernet (1)
- Arrival Time: Jan 24, 2026 04:34:37.393063346 UTC
- UTC Arrival Time: Jan 24, 2026 04:34:37.393063346 UTC
- Epoch Arrival Time: 1769229277.393063346
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 1.065521245 seconds]
- [Time since reference or first frame: 11.062069289 seconds]
- Frame Number: 16
- Frame Length: 81 bytes (648 bits)
- Capture Length: 81 bytes (648 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: eth:ethertype:ip:udp:dns]
- Character encoding: ASCII (0)
- [Coloring Rule Name: UDP]
- [Coloring Rule String: udp]
- Ethernet II, Src: PCSSystemtec_ed:74:19 (08:00:27:ed:74:19), Dst: 52:55:0a:00:02:02 (52:55:0a:00:02:02)
- Destination: 52:55:0a:00:02:02 (52:55:0a:00:02:02)
-1. = LG bit: Locally administered address (this is NOT the factory default)
-0. = TC bit: Individual address (unicast)

The packet bytes pane shows the raw data of the packet, including the Ethernet II header, IP header, and UDP header.

Step 22

DNS request failed

Explanation: Initial DNS server does not respond.

First it tried with 10.255.255.1 and it waited for aprox. 5 seconds and then again tried with original DNS server i:e 192.168.1.1

The image shows a Wireshark packet capture window titled "Capturing from enp0s3". The packet list pane shows several DNS packets. Packet 20 is highlighted, showing a standard query for "mygreatlearning.com" from 10.0.2.15 to 10.125.82.152. The packet details pane shows the encapsulation type as Ethernet II, and the Ethernet II section shows the destination MAC address as 52:55:0a:00:02:02. The packet bytes pane shows the raw data of the packet.

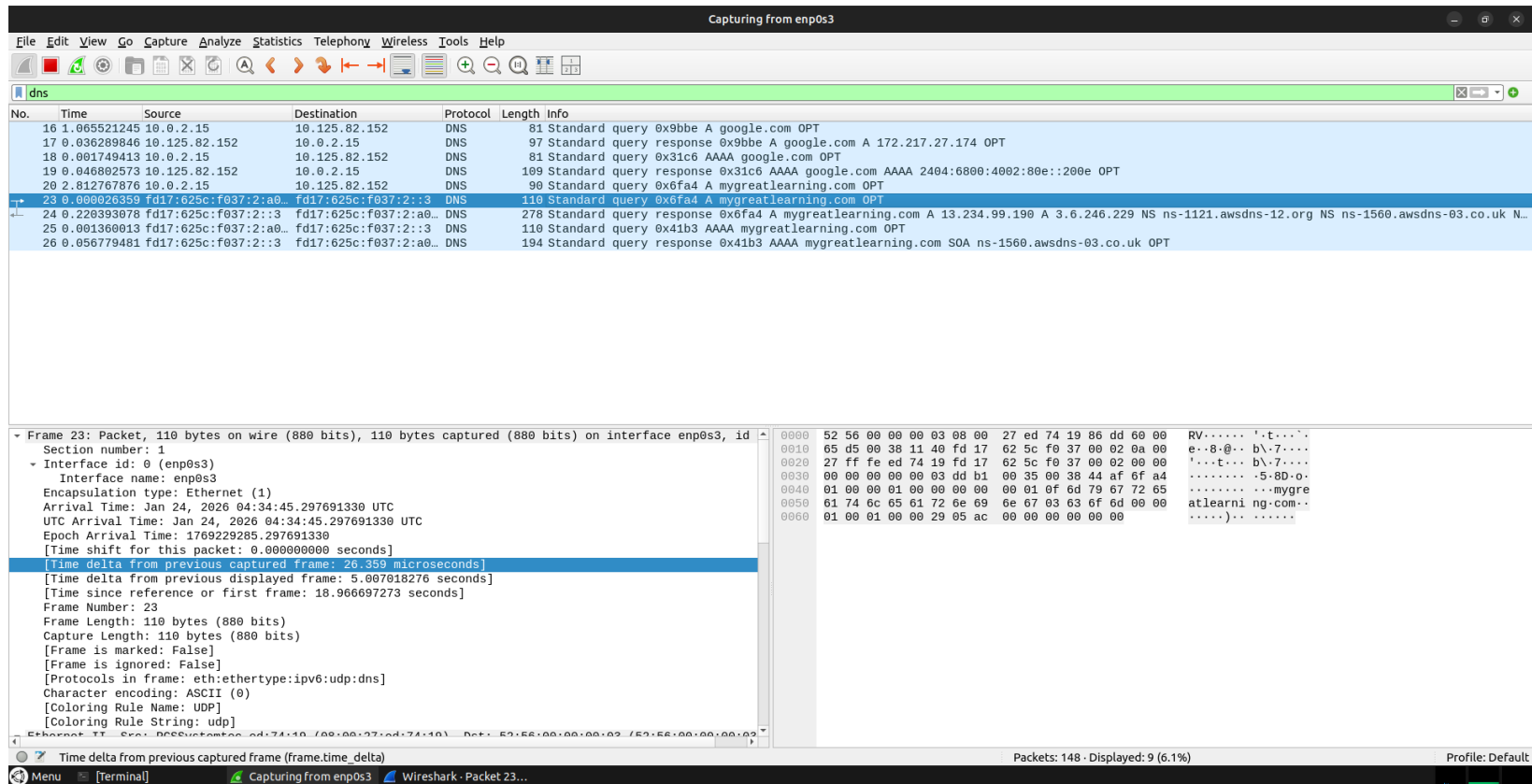
Encapsulation type: Ethernet (1)
Arrival Time: Jan 24, 2026 04:34:40.290673054 UTC
UTC Arrival Time: Jan 24, 2026 04:34:40.290673054 UTC
Epoch Arrival Time: 1769229280.290673054
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 2.812767876 seconds]
[Time delta from previous displayed frame: 2.812767876 seconds]
[Time since reference or first frame: 13.959678997 seconds]
Frame Number: 20
Frame Length: 90 bytes (720 bits)
Capture Length: 90 bytes (720 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:udp:dns]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
Ethernet II, Src: PCSSysntec_ed:74:19 (08:00:27:ed:74:19), Dst: 52:55:0a:00:02:02 (52:55:0a:00:02:02)
Destination: 52:55:0a:00:02:02 (52:55:0a:00:02:02)
.....1..... = LG bit: Locally administered address (this is NOT the factory default)
.....0..... = IG bit: Individual address (unicast)
Source: PCSSysntec_ed:74:19 (08:00:27:ed:74:19)

0000 52 55 0a 00 02 02 08 00 27 ed 74 19 08 00 45 00 RU.....'.t...E-
0010 00 4c 2a 23 00 00 40 11 e7 5a 0a 00 02 0f 0a 7d .L*#..@. .Z.....}
0020 52 98 b0 2a 00 35 00 38 69 6d 6f a4 01 00 00 01 R..*.5-8 imo.....
0030 00 00 00 00 00 01 0f 6d 79 67 72 65 61 74 6c 65m ygreatle
0040 61 72 6e 69 6e 67 03 63 6f 6d 00 00 01 00 01 00 arning.c om.....
0050 00 29 05 c0 00 00 00 00 00 00)
Packets: 144 · Displayed: 9 (6.3%) Profile: Default

Step 23

Retry with original DNS

Explanation: System retries using secondary DNS server i:e 192.168.1.1



File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
16	1.065521245	10.0.2.15	10.125.82.152	DNS	81	Standard query 0x9bbe A google.com OPT
17	0.036289846	10.125.82.152	10.0.2.15	DNS	97	Standard query response 0x9bbe A google.com A 172.217.27.174 OPT
18	0.001749413	10.0.2.15	10.125.82.152	DNS	81	Standard query 0x31c6 AAAA google.com OPT
19	0.046802573	10.125.82.152	10.0.2.15	DNS	109	Standard query response 0x31c6 AAAA google.com AAAA 2404:6800:4002:80e::200e OPT
20	2.812767876	10.0.2.15	10.125.82.152	DNS	90	Standard query 0x6fa4 A mygreatlearning.com OPT
23	0.000026359	fd17:625c:f037:2:a0...	fd17:625c:f037:2::3	DNS	110	Standard query 0x6fa4 A mygreatlearning.com OPT
24	0.220393078	fd17:625c:f037:2::3	fd17:625c:f037:2:a0...	DNS	278	Standard query response 0x6fa4 A mygreatlearning.com A 13.234.99.190 A 3.6.246.229 NS ns-1121.awsdns-12.org NS ns-1560.awsdns-03.co.uk N...
25	0.001360013	fd17:625c:f037:2:a0...	fd17:625c:f037:2::3	DNS	110	Standard query 0x41b3 AAAA mygreatlearning.com OPT
26	0.056779481	fd17:625c:f037:2::3	fd17:625c:f037:2:a0...	DNS	194	Standard query response 0x41b3 AAAA mygreatlearning.com SOA ns-1560.awsdns-03.co.uk OPT

Frame 23: Packet, 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface enp0s3, id ...
Section number: 1
Interface id: 0 (enp0s3)
Interface name: enp0s3
Encapsulation type: Ethernet (1)
Arrival Time: Jan 24, 2026 04:34:45.297691330 UTC
UTC Arrival Time: Jan 24, 2026 04:34:45.297691330 UTC
Epoch Arrival Time: 1769229285.297691330
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 26.359 microseconds]
[Time delta from previous displayed frame: 5.007018276 seconds]
[Time since reference or first frame: 18.966697273 seconds]
Frame Number: 23
Frame Length: 110 bytes (880 bits)
Capture Length: 110 bytes (880 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ipv6:udp:dns]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]

0000 52 56 00 00 00 03 08 00 27 ed 74 19 86 dd 60 00 RV.....'.t...'.
0010 65 d5 00 38 11 40 fd 17 62 5c f0 37 00 02 0a 00 e..8@..b\7....
0020 27 ff fe ed 74 19 fd 17 62 5c f0 37 00 02 00 00 '...t...b\7....
0030 00 00 00 00 03 dd b1 00 35 00 38 44 af 6f a4-5-8D-o-
0040 01 00 00 01 00 00 00 00 00 01 0f 6d 79 67 72 65mygre
0050 61 74 6c 65 61 72 6e 69 6e 67 03 63 6f 6d 00 00 atlearn ng.com..
0060 01 00 01 00 00 29 05 ac 00 00 00 00 00 00)

Time delta from previous captured frame (frame.time_delta)

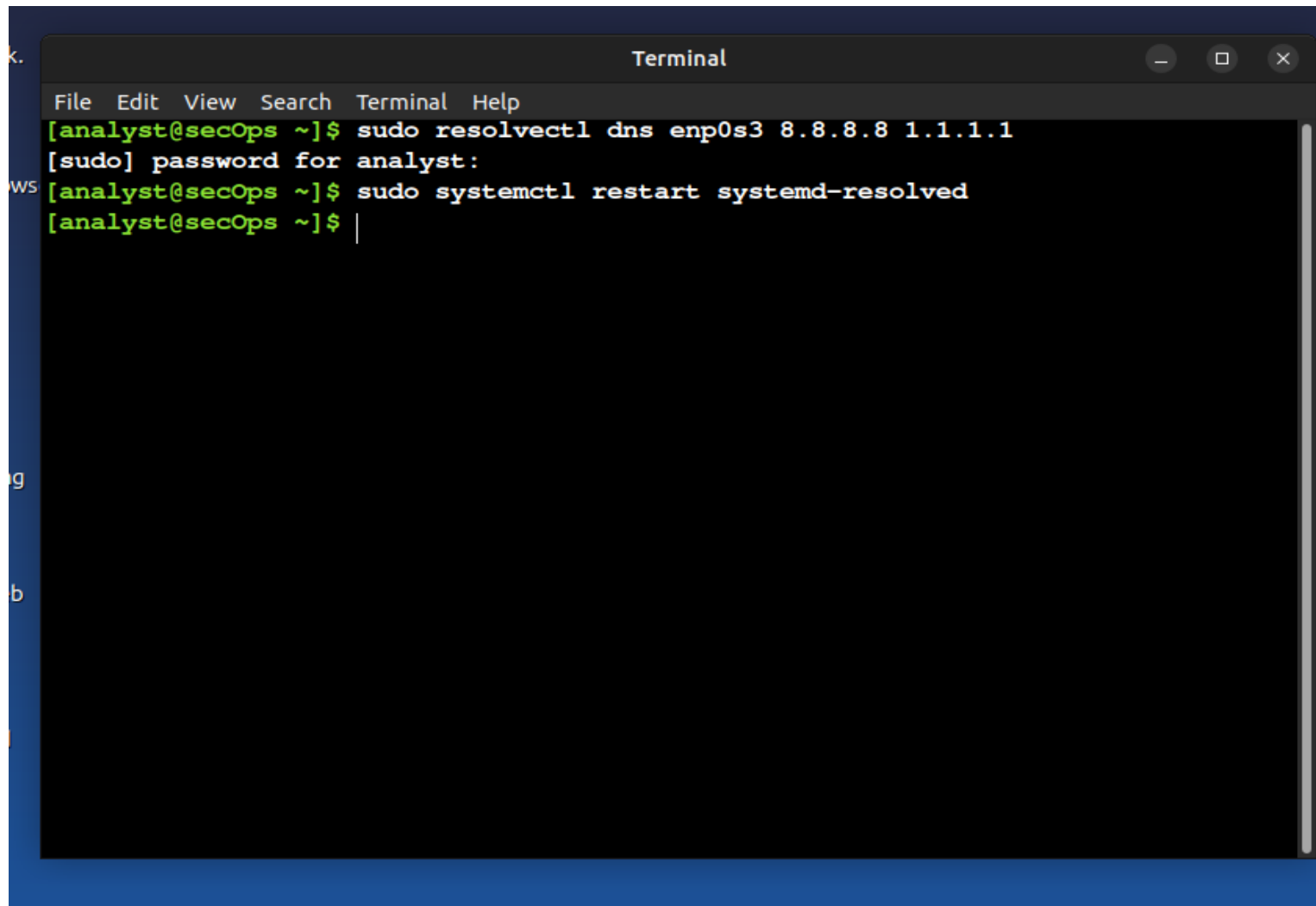
Packets: 148 · Displayed: 9 (6.1%) Profile: Default

Step 24

Command: **sudo resolvectl dns enp0s3 8.8.8.8 1.1.1.1**

Explanation: DNS is changed to public servers (8.8.8.8, 1.1.1.1).

Google = 8.8.8.8 and cloudflare = 1.1.1.1

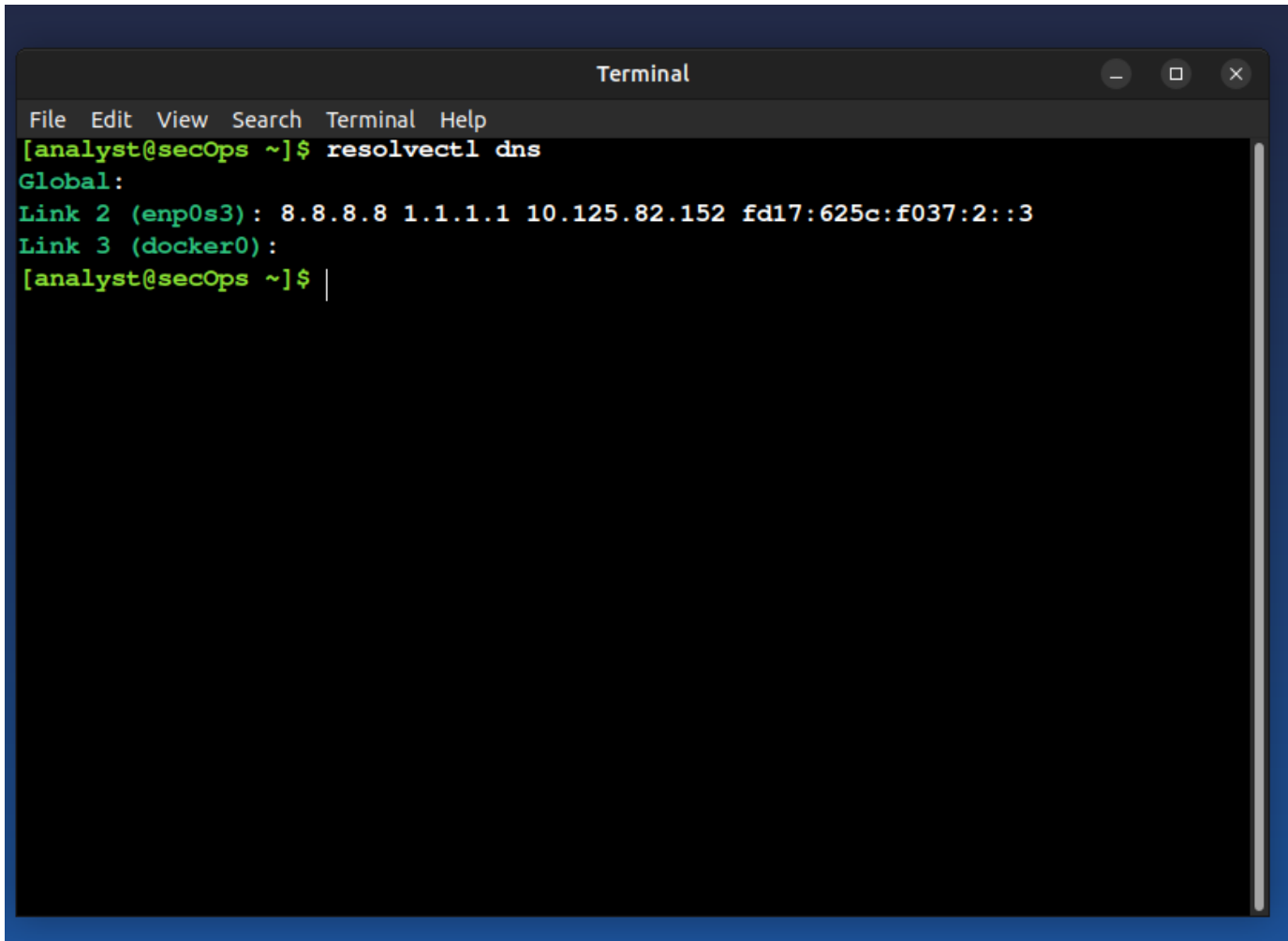
A terminal window titled "Terminal" with a menu bar (File, Edit, View, Search, Terminal, Help) and standard window controls. The terminal shows a user named 'analyst' at a host named 'secOps' in the directory '~'. The user enters the command 'sudo resolvectl dns enp0s3 8.8.8.8 1.1.1.1'. The terminal prompts for a password: '[sudo] password for analyst:'. The user then enters the command 'sudo systemctl restart systemd-resolved'. Finally, the prompt returns to '[analyst@secOps ~]\$' with a cursor. There is a vertical scrollbar on the right side of the terminal window.

```
k.
Terminal
File Edit View Search Terminal Help
[analyst@secOps ~]$ sudo resolvectl dns enp0s3 8.8.8.8 1.1.1.1
[sudo] password for analyst:
ws [analyst@secOps ~]$ sudo systemctl restart systemd-resolved
[analyst@secOps ~]$ |
```

Step 25

Command: **Restart systemd-resolved**

Explanation: Applies the DNS configuration changes.

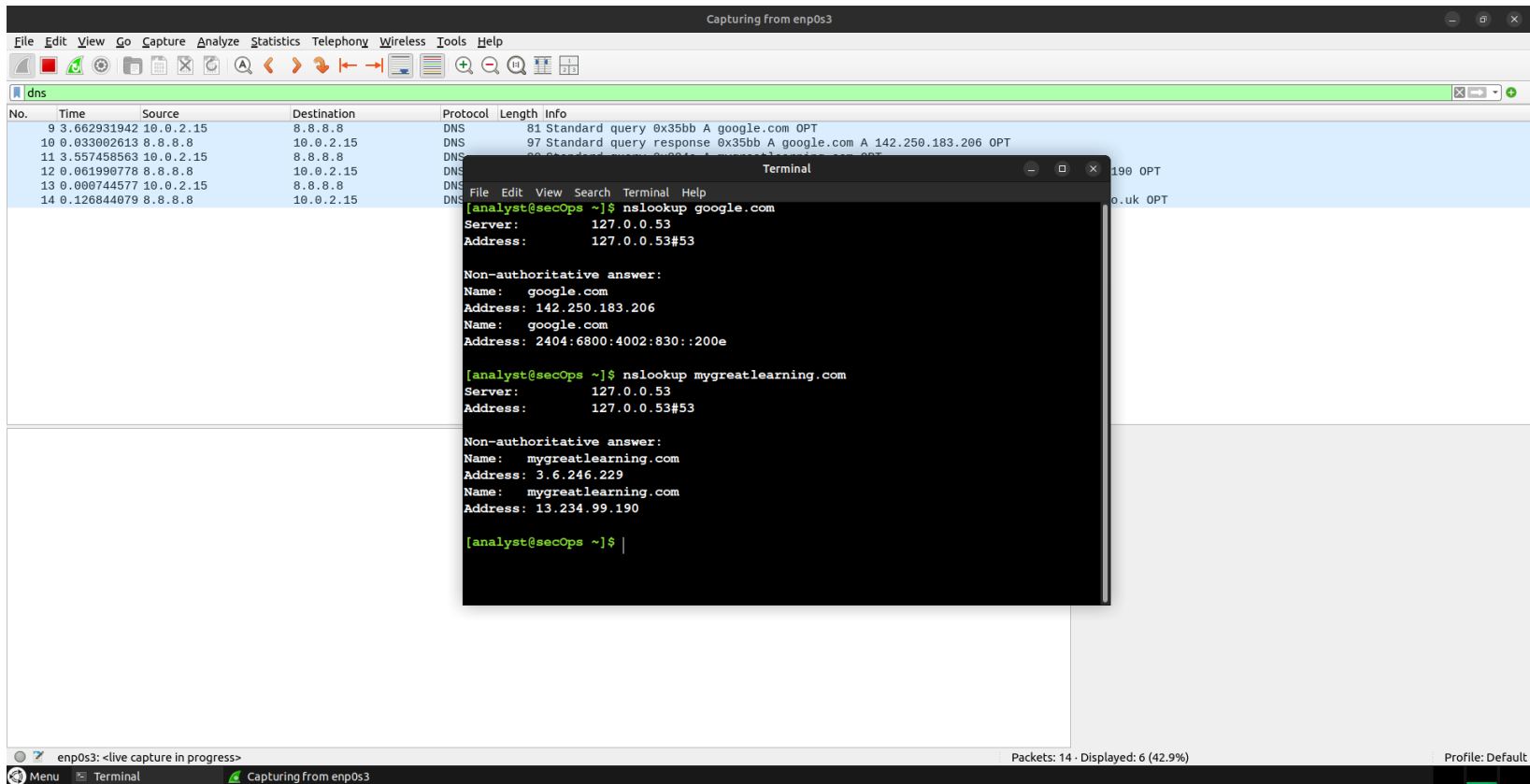
A terminal window titled "Terminal" with standard window controls (minimize, maximize, close) in the top right corner. The menu bar includes "File", "Edit", "View", "Search", "Terminal", and "Help". The terminal content shows a user prompt "[analyst@secOps ~]" followed by the command "resolvectl dns". The output displays the global DNS configuration: "Global:" followed by "Link 2 (enp0s3): 8.8.8.8 1.1.1.1 10.125.82.152 fd17:625c:f037:2::3" and "Link 3 (docker0):". The prompt returns to "[analyst@secOps ~]" with a cursor at the end.

```
Terminal
File Edit View Search Terminal Help
[analyst@secOps ~]$ resolvectl dns
Global:
Link 2 (enp0s3): 8.8.8.8 1.1.1.1 10.125.82.152 fd17:625c:f037:2::3
Link 3 (docker0):
[analyst@secOps ~]$
```

Step 26

Test DNS after fix

Explanation: DNS resolution becomes fast again.



The screenshot displays a Wireshark network capture window titled "Capturing from enp0s3" with the filter set to "dns". The packet list shows several DNS queries and responses. A terminal window is overlaid on the Wireshark interface, showing the output of two `nslookup` commands.

Wireshark Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
9	3.662931942	10.0.2.15	8.8.8.8	DNS	81	Standard query 0x35bb A google.com OPT
10	0.033002613	8.8.8.8	10.0.2.15	DNS	97	Standard query response 0x35bb A google.com A 142.250.183.206 OPT
11	3.557458563	10.0.2.15	8.8.8.8	DNS	80	Standard query 0x35bb A mygreatlearning.com OPT
12	0.061990778	8.8.8.8	10.0.2.15	DNS	97	Standard query response 0x35bb A mygreatlearning.com A 3.6.246.229 OPT
13	0.000744577	10.0.2.15	8.8.8.8	DNS	80	Standard query 0x35bb A mygreatlearning.com OPT
14	0.126844079	8.8.8.8	10.0.2.15	DNS	97	Standard query response 0x35bb A mygreatlearning.com A 13.234.99.190 OPT

Terminal Output:

```
[analyst@secOps ~]$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.183.206
Name:   google.com
Address: 2404:6800:4002:830::200e

[analyst@secOps ~]$ nslookup mygreatlearning.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   mygreatlearning.com
Address: 3.6.246.229
Name:   mygreatlearning.com
Address: 13.234.99.190

[analyst@secOps ~]$
```

The terminal output confirms that DNS resolution is now fast, as evidenced by the immediate responses for both `google.com` and `mygreatlearning.com`.

Step 27

Verify timing

Explanation: Wireshark confirms normal DNS response time.

The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains various icons for file operations, capture control, and analysis. The packet list pane shows a list of captured packets, with the selected packet (No. 10) highlighted in blue. The packet details pane shows the structure of the selected packet, including the Ethernet II header, UDP header, and DNS header. The packet bytes pane shows the raw data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
9	3.662931942	10.0.2.15	8.8.8.8	DNS	81	Standard query 0x35bb A google.com OPT
10	0.033002613	8.8.8.8	10.0.2.15	DNS	97	Standard query response 0x35bb A google.com A 142.250.183.206 OPT
11	3.557458563	10.0.2.15	8.8.8.8	DNS	90	Standard query 0x804e A mygreatlearning.com OPT
12	0.061990778	8.8.8.8	10.0.2.15	DNS	122	Standard query response 0x804e A mygreatlearning.com A 3.6.246.229 A 13.234.99.190 OPT
13	0.000744577	10.0.2.15	8.8.8.8	DNS	90	Standard query 0x5642 AAAA mygreatlearning.com OPT
14	0.126844079	8.8.8.8	10.0.2.15	DNS	174	Standard query response 0x5642 AAAA mygreatlearning.com SOA ns-1560.awsdns-03.co.uk OPT

Frame 10: Packet, 97 bytes on wire (776 bits), 97 bytes captured (776 bits) on interface enp0s3, id 0

- Section number: 1
- Interface id: 0 (enp0s3)
- Interface name: enp0s3
- Encapsulation type: Ethernet (1)
- Arrival Time: Jan 24, 2026 04:47:15.365939741 UTC
- UTC Arrival Time: Jan 24, 2026 04:47:15.365939741 UTC
- Epoch Arrival Time: 1769230035.365939741
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 33.002613 milliseconds]
- [Time delta from previous displayed frame: 33.002613 milliseconds]
- [Time since reference or first frame: 6.038395726 seconds]
- Frame Number: 10
- Frame Length: 97 bytes (776 bits)
- Capture Length: 97 bytes (776 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: eth:ethertype:ip:udp:dns]
- Character encoding: ASCII (0)
- [Coloring Rule Name: UDP]
- [Coloring Rule String: udp]

Ethernet II, Src: 52:55:0a:00:02:02 (52:55:0a:00:02:02), Dst: PCSystemtec_ed:74:19 (08:00:27:ed:74:19)

Time delta from previous captured frame (frame.time_delta)

Packets: 18 · Displayed: 6 (33.3%) Profile: Default